

# A Global Spectral Gap for MALA with a Uniformly Randomized Step Size

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## Abstract

Let  $\pi(dx) \propto e^{-U(x)} dx$  on  $\mathbb{R}^d$ , where  $0 < m \leq L < \infty$ ,  $mI_d \preceq \nabla^2 U(x) \preceq LI_d$ , and  $\kappa = L/m$ . It is known that, under warm-start assumptions, fixed-step Metropolis-adjusted Langevin algorithm (MALA) with properly tuned step size has mixing time of order  $\kappa\sqrt{d}$  up to logarithmic factors. By contrast, when the condition number is bounded away from one, no single fixed step size yields a matching spectral-gap lower bound of order  $(\kappa\sqrt{d})^{-1}$  uniformly over this target class. We show that MALA with a uniformly randomized step size admits a spectral-gap lower bound of this size. At each iteration, the randomized-step MALA considered here draws  $h$  uniformly from  $(0, H)$  and performs one ordinary MALA transition with step size  $h$ . We show that, when  $H$  is of order  $(L\sqrt{d})^{-1}$ , the right spectral gap of randomized-step MALA admits a lower bound of order

$$\frac{1}{\kappa\sqrt{d}[1 + \log(d+1) + \log \kappa]}.$$

The main new ingredient in the proof is a Cheeger-type inequality for aggregating estimates of the one-step flow of MALA out of measurable sets at various step-size scales. It allows the scale used to control the flow to depend on the set and avoids the additional loss that would result from first summing the flows and then applying the standard Cheeger inequality.

## 1 Introduction

For a target probability distribution  $\pi$  on  $\mathbb{R}^d$  with density proportional to  $e^{-U}$ , the Metropolis-adjusted Langevin algorithm (MALA) proposes

$$Y = x - h\nabla U(x) + \sqrt{2h}Z, \text{ where } Z \sim N_d(0, I_d),$$

and applies the Metropolis–Hastings correction. MALA is one of the most fundamental Markov chain Monte Carlo (MCMC) algorithms and has been studied extensively. Its classical geometric properties and high-dimensional scaling were developed by [Roberts and Tweedie \[1996\]](#), [Roberts and Rosenthal \[1998\]](#); it is also closely related to Hamiltonian Monte Carlo [[Neal, 2011](#), [Wu et al., 2022](#), [Chen and Gtarmiry, 2023](#)].

For an  $m$ -strongly convex and  $L$ -smooth potential  $U(\cdot)$ , [Wu et al. \[2022\]](#), strengthening results in [Chewi et al. \[2021\]](#), show that the mixing time of (lazy) MALA from a warm start has the minimax  $\kappa\sqrt{d}$  dependence, up to logarithmic factors, where  $\kappa = L/m$  is the condition number. This rate is achieved with a MALA step size  $h$  of order  $(L\sqrt{d})^{-1}$  up to logarithmic factors. The proofs in [Chewi et al. \[2021\]](#) and [Wu et al. \[2022\]](#) control acceptance on a high-probability region and then use an  $s$ -conductance argument [[Lovász and Simonovits, 1993](#)]. They therefore do not lower-bound

the ordinary conductance over sets of every positive stationary mass, and their arguments do not yield a matching global Poincaré bound of the form

$$\text{Gap}_R(P_h) \gtrsim \frac{1}{\kappa\sqrt{d}}, \quad (1)$$

where  $P_h$  is the Markov kernel of MALA with step size  $h$ ,  $\text{Gap}_R(\cdot)$  denotes the right spectral gap, and  $\gtrsim$  suppresses universal constants and the logarithmic factors. (Recall that, for a lazy reversible Markov chain, the mixing time is roughly inversely proportional to the right spectral gap, so a gap of order  $(\kappa\sqrt{d})^{-1}$  implies mixing time of order  $\kappa\sqrt{d}$ .) Under only strong convexity and gradient Lipschitzness, a fixed-step global Poincaré bound at the  $d^{-1/2}$  scale is impossible uniformly over the target class. Indeed, following arguments from [Chewi et al. \[2021\]](#), [Wu et al. \[2022\]](#), we show that a bound of the form (1) cannot hold uniformly over all  $m$ -strongly convex and  $L$ -smooth potentials if  $\kappa - 1$  is greater than a positive constant. In fact, in [Proposition 2.3](#), it is shown that, when  $\kappa \geq \kappa_0 > 1$  for some constant  $\kappa_0$ , for any  $h > 0$ , there exists an  $m$ -strongly convex and  $L$ -smooth potential  $U$  such that  $\text{Gap}_R(P_h)$  is upper bounded by a  $\kappa_0$ -dependent multiple of the maximum of  $\log(\kappa d)/(\kappa d)$  and  $e^{-cd}$ , where  $c > 0$  is universal.

A global spectral-gap bound is useful for more than deriving a mixing-time estimate. Let  $P$  be a lazy,  $\pi$ -reversible Markov kernel. If  $\mu$  is an initial distribution with  $d\mu/d\pi \in L^2(\pi)$ , then

$$\|\mu P^n - \pi\|_{\text{TV}} \leq \frac{1}{2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} [1 - \text{Gap}_R(P)]^n.$$

Thus a spectral gap certifies convergence all the way to stationarity, rather than only for a prescribed accuracy [[Kontoyiannis and Meyn, 2012](#), [Roberts and Rosenthal, 1997](#), [Rudolf, 2012](#)]. More importantly for Monte Carlo, for a stationary reversible chain and a centered function in  $L^2(\pi)$ , a positive right spectral gap implies a central limit theorem for Markov-chain averages and controls their asymptotic variance [[Kipnis and Varadhan, 1986](#)]. Related spectral arguments also yield finite-sample variance, mean-square error, and concentration bounds for Markov-chain averages [[Rudolf, 2012](#), [Paulin, 2015](#)]. By contrast, a favorable warm-start mixing bound can coexist with very high autocorrelation for certain  $L^2(\pi)$  test functions.

Quantitative global spectral-gap results for MALA are considerably sparser than mixing-time bounds. The classical analysis of [Roberts and Tweedie \[1996\]](#) gives qualitative criteria for geometric ergodicity, rather than dimension-explicit lower bounds on the gap, while [Bou-Rabee and Hairer \[2013\]](#) show that, for drifts that are not globally Lipschitz, MALA may fail to have a spectral gap even when the underlying Langevin diffusion does. [Wu et al. \[2022\]](#) use spectral-gap upper bounds of order  $mh$  for specially constructed targets to establish their minimax mixing-time lower bound, but the existing fixed-step analyses do not provide a matching global spectral-gap lower bound when  $h$  is of order  $(L\sqrt{d})^{-1}$ . More recently, [Liu and Tong \[2026\]](#) establish  $\text{Gap}_R(P_h) \gtrsim (\kappa^2 d)^{-1}$  when  $h \lesssim (\kappa L d)^{-1}$ . This spectral-gap bound gives a mixing-time bound of order  $\kappa^2 d$ , which is significantly worse than the  $\kappa\sqrt{d}$  dependence achieved with a step size of order  $(L\sqrt{d})^{-1}$ .

We study the randomized kernel

$$\bar{P}_H = \frac{1}{H} \int_0^H P_h \, dh.$$

Operationally, the algorithm samples  $h \sim \text{Unif}(0, H)$  and then performs one ordinary MALA step. It uses one proposal and has essentially the same computational cost as fixed-step MALA. By linearity of detailed balance, it is exactly  $\pi$ -reversible. In [Theorem 2.1](#), we show that if  $U$  is twice continuously differentiable and  $H$  is of order  $(L\sqrt{d})^{-1}$ , then  $\text{Gap}_R(\bar{P}_H)$  has a lower bound of order  $(\kappa\sqrt{d})^{-1}$ , up to logarithmic factors, matching the minimax mixing-time bound for fixed-step MALA.

An accompanying Lean 4 development formally verifies the main theorem and the other principal theorem-level conclusions from their stated assumptions. This includes the randomized-step lower bound and its lazy and square-root dimension consequences, the fixed-step minimax obstruction, the all-parameter flow estimates, and the fractional and hard-assignment aggregation results. The scope of the verification, including the alternative discrete proof used in place of the continuous-time argument in Appendix B, is described in [Section 5](#).

Randomization of Metropolis tuning parameters has been advocated as a way to improve robustness. In the setting most directly relevant here, [Grazzi et al. \[2026\]](#) study the randomized kernel  $\int P_h \nu(dh)$ , where  $\nu$  is a distribution on  $(0, \infty)$ . They show that randomized kernels inherit spectral gaps and weak Poincaré inequalities from suitable fixed-step components and that randomization can turn severe sensitivity to step-size misspecification into a much milder deterioration of the spectral gap. The intuition is that a randomized method may sacrifice some peak asymptotic efficiency in exchange for performing reasonably over a much wider range of scales. Related benefits of randomization are known for Hamiltonian methods [[Bou-Rabee and Sanz-Serna, 2017](#), [Apers et al., 2024](#)].

As shown in [Grazzi et al. \[2026\]](#), the randomized kernel satisfies the inheritance bound  $\text{Gap}_R(\bar{P}_H) \geq H^{-1} \int_0^H \text{Gap}_R(P_h) dh$ . This inequality alone, however, does not yield the  $(\kappa\sqrt{d})^{-1}$  rate sought here, because no available fixed-step result establishes  $\text{Gap}_R(P_h) \gtrsim (\kappa\sqrt{d})^{-1}$ , regardless of  $h$ . To be more specific, existing conductance lower bounds, which are translated into spectral-gap lower bounds via Cheeger’s inequality [[Lawler and Sokal, 1988](#)], are considerably smaller than desired at every scale. Recall that Cheeger’s inequality asserts that  $\text{Gap}_R(P_h)$  is lower bounded by  $\phi(P_h)^2/2$ , where  $\phi(P_h)$  is the global conductance:

$$\phi(P_h) := \inf_{S: 0 < \pi(S) \leq 1/2} \frac{J_{P_h}(S, S^c)}{\pi(S)}, \quad J_{P_h}(S, S^c) = \int_S P_h(x, S^c) \pi(dx).$$

To apply Cheeger’s inequality, we want a lower bound on  $\phi(P_h)$  that is of order  $(\kappa\sqrt{d})^{-1/2}$  for a sufficiently large set of  $h \in [0, H]$ . More precisely, [Liu and Tong \[2026\]](#) obtain a lower bound on  $\phi(P_h)$  of order  $(\kappa^2 d)^{-1/2}$  for  $h \lesssim (\kappa L d)^{-1}$ . For  $h$  of order  $(L\sqrt{d})^{-1}$ , [Wu et al. \[2022\]](#) obtain a lower bound of order  $(\kappa\sqrt{d})^{-1/2}$  on  $J_{P_h}(S, S^c)/\pi(S)$  when  $\pi(S)$  exceeds a certain threshold; below that threshold, however, the ratio can scale very poorly with  $d$ .

Our argument exploits information that the preceding inheritance bound discards: the step-size range used to control the flow out of a set may depend on the stationary mass of that set. For each measurable  $S \subset \mathbb{R}^d$ , we select a range of step sizes tailored to  $\pi(S)$ , using progressively smaller steps as  $\pi(S)$  decreases. When  $\pi(S)$  is not too small, comparatively larger steps can be used: the Metropolis acceptance rate is lower bounded by a positive constant over  $S$ , outside an exceptional set with small stationary mass [[Chewi et al., 2021](#), [Wu et al., 2022](#)]. Writing  $\mathcal{K}_t = (2/t) \int_{t/2}^t P_h dh$  for the kernel averaged over the upper dyadic half of  $(0, t)$ , one can then lower-bound the one-step flow  $J_{\mathcal{K}_t}(S, S^c)$  by a suitable multiple of  $\pi(S)$ . For smaller sets, the exceptional region in the preceding argument can no longer be ignored, and we instead use a smaller, globally safe step size which leads to a constant uniform lower bound on the acceptance rate over the whole space. Gaussian isoperimetry then supplies the corresponding lower bound on the flow out of  $S$ .

Finally, we combine the flow estimates obtained at the different scales using a Cheeger-type aggregation lemma ([Lemma 3.5](#)) for mixtures of reversible Markov kernels. The lemma lets us use relatively large steps for sets of moderate stationary mass and smaller steps for rarer sets, without losing an additional factor equal to the probability of selecting the useful step-size range. The closest analytic predecessor is the general Cheeger inequality of [Chen and Wang \[2000\]](#), which can be used to derive [Lemma 3.5](#). We give a direct proof and a more transparent formulation ([Theorem 3.6](#)) suitable for the present setting.

## Relation to existing work

Controlling the rejection rate is central to the analysis of MALA. There are several approaches to controlling MALA rejection at the  $d^{-1/2}$  scale. [Chewi et al. \[2021\]](#) use a projection characterization of Metropolis adjustment and compare the Euler proposal with exact Langevin diffusion. [Wu et al. \[2022\]](#) instead compare one leapfrog step with continuous Hamiltonian dynamics. Earlier pathwise comparisons between Metropolized integrators and their limiting diffusions include [Bou-Rabee and Vanden-Eijnden \[2010\]](#), [Bou-Rabee and Hairer \[2013\]](#).

Our appendix follows the diffusion-comparison philosophy but uses a different route tailored to the stationary, pointwise rejection function. The stationary increment bound follows from the elementary  $L^2$  conjugation estimate that is the  $p = 2$  energy core of Davies’ exponential-perturbation method [[Davies, 1989](#), Section 3.2, especially Lemma 3.2.1 and p. 85]; the estimate is proved directly in the appendix below. See also [Ouhabaz \[2005, Chapter 6\]](#). The same increment bound can alternatively be obtained from the Lyons–Zheng forward–backward decomposition [[Fukushima et al., 2011](#), Theorem 5.7.1]; see also [Lyons and Zheng \[1988\]](#). A Girsanov change of path law then compares the exact diffusion with the drift frozen at its initial state. Endpoint conditioning and the Metropolis–Hastings meet identity [[Tierney, 1998](#), [Billera and Diaconis, 2001](#)] convert the path estimate into an  $L^p(\pi)$  rejection bound. Quantitative martingale inequalities introduce an additional factor  $\sqrt{p}$ ; this is the source of the final logarithmic loss.

The close-coupling and isoperimetric part of the argument is standard in modern Metropolis analyses [[Dwivedi et al., 2019](#), [Wu et al., 2022](#), [Andrieu et al., 2024](#), [Liu and Tong, 2026](#)]. The use of a scale indexed by set size is related in spirit to average conductance, evolving sets, blocking conductance, and spectral profiles [[Lovász and Kannan, 1999](#), [Morris and Peres, 2005](#), [Goel et al., 2006](#), [Kannan et al., 2006](#)].

The closest analytic predecessor of our aggregation lemma is [Chen and Wang \[2000\]](#). They prove a general version of Cheeger’s inequality that applies in settings considerably broader than the Markov-kernel setting considered here and extends the earlier results of [Lawler and Sokal \[1988\]](#). Their result can be used to derive our aggregation lemma, although the proof we give is direct and self-contained. The part specific to our argument is the way the lemma is combined with the set-dependent choice of scale: different sets may require different step-size ranges. We use this idea to combine flow estimates over a geometric sequence of step-size ranges for randomized-step MALA.

## Generative AI disclosure

The authors used GPT-5.6 Sol through ChatGPT to assist with drafting and revising the major proofs in this article. The authors checked and edited every argument and take full responsibility for the accuracy and content of the manuscript.

## 2 Setup and Main Result

Assume throughout that  $d$  is a positive integer and that  $U : \mathbb{R}^d \rightarrow \mathbb{R}$  is twice continuously differentiable with

$$mI_d \preceq \nabla^2 U(x) \preceq LI_d, \quad x \in \mathbb{R}^d,$$

where  $0 < m \leq L < \infty$ . Let  $\kappa := L/m \geq 1$ , and define the target distribution

$$\pi(dx) = \frac{e^{-U(x)}}{\int_{\mathbb{R}^d} e^{-U(u)} du} dx.$$

For  $h > 0$ , let  $Q_h(x, \cdot)$  be the Gaussian law

$$N_d(x - h\nabla U(x), 2hI_d).$$

The fixed-step MALA kernel is

$$P_h(x, dy) = \alpha_h(x, y) Q_h(x, dy) + \left[1 - \int \alpha_h(x, z) Q_h(x, dz)\right] \delta_x(dy),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}$$

and  $q_h$  is the density of  $Q_h$ . The usual Metropolis–Hastings argument makes  $P_h$   $\pi$ -reversible [Tierney, 1998]; consequently, so is the randomized-step MALA kernel  $\bar{P}_H = H^{-1} \int_0^H P_h dh$ .

For probability measures  $\mu, \nu$  on a measurable space  $(\Omega, \mathcal{F})$ , write

$$\text{TV}(\mu, \nu) = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|.$$

For a probability measure  $\varpi$ , let  $L^2(\varpi)$  be the space of measurable functions  $f : \Omega \rightarrow \mathbb{R}$  such that  $\int f^2 d\varpi < \infty$ . For a  $\varpi$ -reversible kernel  $P$  and  $f \in L^2(\varpi)$ , define the Dirichlet form

$$\mathcal{E}_P(f, f) = \frac{1}{2} \int [f(x) - f(y)]^2 \varpi(dx) P(x, dy).$$

The right, or Poincaré, spectral gap is defined as

$$\text{Gap}_R(P) = \inf_{\substack{f \in L^2(\varpi) \\ \text{Var}_\varpi(f) > 0}} \frac{\mathcal{E}_P(f, f)}{\text{Var}_\varpi(f)}.$$

This quantity controls the convergence properties of the Markov chain associated with  $P$  if  $P$  is lazy or positive semidefinite.

The main result of this paper is stated below.

**Theorem 2.1.** *There are universal constants  $A_0 \in [2, \infty)$ ,  $b_0 \in (0, 1/2]$ , and  $c_0 \in (0, \infty)$  such that the following holds. Define*

$$p_\star = A_0 \{1 + \log(d+1) + \log \kappa\}.$$

*Then, for every  $H > 0$ ,*

$$\text{Gap}_R(\bar{P}_H) \geq c_0 \frac{m}{H} \min \left\{ H, \frac{b_0}{L} \max \left\{ \frac{1}{\sqrt{p_\star(d+p_\star)}}, \frac{1}{d} \right\} \right\}^2. \quad (2)$$

*For the lazy kernel  $(I + \bar{P}_H)/2$ , the right-hand side is divided by two.*

**Corollary 2.2.** *Under the assumptions of Theorem 2.1, let  $H = c/(L\sqrt{d})$  for some  $c > 0$ . Then*

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{b_0^2}{c} \max \left\{ \frac{d}{p_\star(d+p_\star)}, \frac{1}{d} \right\} \right\}.$$

*In particular, since  $\min\{p_\star(d+p_\star)/d, d\} \leq 2p_\star$ ,*

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{b_0^2}{2cp_\star} \right\}.$$

The spectral gap lower bound of order  $(\kappa\sqrt{d})^{-1}$  for randomized-step MALA is in sharp contrast with the following upper bound on the spectral gap of fixed-step MALA, which is proved in [Appendix A](#).

**Proposition 2.3.** *Fix  $\kappa_0 > 1$ . There are a universal constant  $c > 0$  and a constant  $C > 0$  depending only on  $\kappa_0$  such that the following holds. Let  $d \geq 2$ ,  $0 < m < L$ , and  $\kappa = L/m \geq \kappa_0$ . Then*

$$\sup_{h>0} \inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \leq \nabla^2 U(x) \leq LI_d, \ x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq C \max \left\{ \frac{\log(\kappa d)}{\kappa d}, e^{-cd} \right\}. \quad (3)$$

Here  $P_{U,h}$  denotes the MALA kernel with step size  $h$  and target density proportional to  $e^{-U}$ .

The proof of [Proposition 2.3](#) utilizes constructions from [Chewi et al. \[2021\]](#) and [Wu et al. \[2022\]](#). Both of these works construct a hard target distribution and use it to bound the spectral gap of fixed-step MALA from above. Their results do not readily yield the upper bound herein. We use a modified version of the construction in [Wu et al. \[2022\]](#) to obtain the bound.

### 3 Ingredients of the Proof

We now describe the main ingredients of the proof of [Theorem 2.1](#). In [Proposition B.1](#), we bound the MALA rejection rate and use this bound to establish the close-coupling result in [Proposition 3.2](#). The curvature assumption on  $U$  yields the isoperimetric inequality in [Proposition 3.3](#). We then combine [Propositions 3.2](#) and [3.3](#) in [Proposition 3.4](#) to bound the one-step MALA flow at the appropriate scales. Importantly, for each set  $S$ , the flow out of  $S$  is controlled for step sizes in a subinterval of  $[0, H]$  that depends on  $S$ . The general component-aggregation bound in [Theorem 3.6](#) combines the one-step flows at various scales, yielding [Theorem 2.1](#).

For  $t > 0$ , define the MALA kernel averaged over the upper dyadic half of  $(0, t)$  by

$$\mathcal{K}_t = \frac{2}{t} \int_{t/2}^t P_h \, dh.$$

The uniform mixture  $\bar{P}_H$  contains this component with weight  $t/(2H)$  whenever  $t \leq H$ . The following elementary relation holds.

**Lemma 3.1.** *Let  $0 < t_J < \dots < t_0 \leq H$ , and suppose that the intervals  $(t_j/2, t_j)$ ,  $j \in \{0, \dots, J\}$ , are pairwise disjoint. Then, for  $f \in L^2(\pi)$ ,*

$$\mathcal{E}_{\bar{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f).$$

*Proof.* By nonnegativity and Tonelli's theorem,

$$\mathcal{E}_{\bar{P}_H}(f, f) = \frac{1}{H} \int_0^H \mathcal{E}_{P_h}(f, f) \, dh \geq \sum_{j=0}^J \frac{1}{H} \int_{t_j/2}^{t_j} \mathcal{E}_{P_h}(f, f) \, dh.$$

The definition of  $\mathcal{K}_{t_j}$  gives the claim. □

### 3.1 Close coupling

The next proposition, proved in [Appendix B](#), shows that  $\mathcal{K}_t(x, \cdot)$  and  $\mathcal{K}_t(y, \cdot)$  are close in total variation whenever  $\|x - y\| \leq \sqrt{t}/16$ . For the larger admissible values of  $t$ , this conclusion is guaranteed only when  $x$  and  $y$  lie outside a small exceptional set.

**Proposition 3.2.** *There are universal constants  $c_r, C_r > 0$  such that, for every real  $p \geq 2$  and every  $t$  satisfying*

$$0 < t \leq \frac{c_r}{L\sqrt{p(d+p)}}, \quad (4)$$

*there is a measurable set  $G_{t,p} \subseteq \mathbb{R}^d$  satisfying*

$$\pi(G_{t,p}^c) \leq \left[ C_r L t \sqrt{p(d+p)} \right]^p, \quad (5)$$

*and*

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

*Moreover, if  $t' \in (0, 1/(2Ld)]$ , then*

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

### 3.2 Isoperimetric inequality

It is well known that a strongly log-concave distribution satisfies a Gaussian isoperimetric inequality. Corollary 2.2 and equations (2.10)–(2.12) of [Bakry and Ledoux \[1996, pp. 267–268\]](#) imply

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{m}r), \quad (6)$$

where, for  $r > 0$ ,  $A^r = \{x : \text{dist}(x, A) < r\}$ , and  $\Phi$  is the standard Gaussian distribution function. The following three-set form of the isoperimetric inequality is proved in [Appendix C](#).

**Proposition 3.3.** *Let  $A, B, C$  be a measurable partition of  $\mathbb{R}^d$ , let  $r \geq 0$ , and suppose  $\text{dist}(A, B) \geq r$ . If  $0 < q = \min\{\pi(A), \pi(B)\} \leq 1/2$ , then*

$$\pi(C) \geq \frac{q}{4} \min \left\{ 1, r\sqrt{m \log(1/q)} \right\}. \quad (7)$$

### 3.3 One-step flow

For a  $\pi$ -reversible Markov kernel  $K$ , define the one-step flow

$$J_K(S, S^c) = \int_S K(x, S^c) \pi(dx).$$

Combining [Proposition 3.2](#) and [Proposition 3.3](#), we establish the following result concerning  $\mathcal{K}_t$  in [Appendix D](#).

**Proposition 3.4.** *One can find universal constants  $b_0 \in (0, 1/2]$  and  $A_0 \in [2, \infty)$  such that the following holds. Let  $p \geq p_\star := A_0\{1 + \log(d+1) + \log \kappa\}$ ,  $\theta \in (0, 1]$ , and*

$$t = \frac{\theta b_0}{L\sqrt{p(d+p)}}.$$



Then, for every measurable set  $S \subseteq \mathbb{R}^d$  such that  $e^{-p/2} \leq \pi(S) \leq 1/2$ ,

$$J_{\mathcal{K}_t}(S, S^c) \geq 2^{-13} \pi(S) \sqrt{mt \log(1/\pi(S))},$$

where  $mt \log(1/\pi(S)) \leq 1$ .

Moreover, for  $t' \in (0, 1/(2Ld)]$  and every measurable set  $S \subseteq \mathbb{R}^d$  such that  $0 < \pi(S) \leq 1/2$ ,

$$J_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt' \log(1/\pi(S))} \right\}.$$

### 3.4 Aggregation of one-step flows

Cheeger's inequality [Lawler and Sokal, 1988] implies that the spectral gap of a reversible Markov kernel can be estimated by considering the kernel's one-step flow out of measurable sets. It turns out that a direct estimate of the total flow followed by the usual Cheeger inequality loses too much and does not yield the spectral lower bound of order  $(\kappa\sqrt{d})^{-1}$  that we desire. We carry out the more careful approach below.

In Section 4, we partition  $[0, H]$  into disjoint intervals of the form  $[t_j/2, t_j]$ , where  $(t_0, \dots, t_J)$  is a decreasing sequence of positive numbers. For a given measurable set  $S$ , the flow  $J_{\mathcal{K}_{t_j}}(S, S^c)$  may be large for some values of  $j$  but small for others. We therefore assign to each set  $S$  an appropriate index  $j(S)$  and establish a lower bound on  $J_{\mathcal{K}_{t_{j(S)}}}(S, S^c)$ . The purpose of this subsection is to provide the tools for combining these set-dependent bounds into a lower bound on  $\text{Gap}_R(\bar{P}_H)$ .

The following lemma carries out this aggregation for a generic mixture of reversible Markov kernels. Roughly speaking, it allows the kernel that supplies the useful one-step flow bound to depend on the set under consideration, while keeping track of how often each kernel is selected in the mixture. The lemma can also be derived from the general Cheeger inequality of Chen and Wang [2000]. We give a direct proof in Appendix E.

**Lemma 3.5.** *Let  $P, K_1, \dots, K_N$  be Markov kernels that are reversible with respect to a distribution  $\varpi$ . Suppose that there exist constants  $\gamma_1, \dots, \gamma_N > 0$  such that*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \quad \text{for every } f \in L^2(\varpi).$$

*Suppose further that there exist constants  $\beta_1, \dots, \beta_N \geq 0$  satisfying*

$$\sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} > 0$$

*such that, for every measurable set  $S$  satisfying  $0 < \varpi(S) \leq 1/2$ ,*

$$\sum_{j=1}^N \beta_j J_{K_j}(S, S^c) \geq \varpi(S),$$

*where*

$$J_{K_j}(S, S^c) = \int_S K_j(x, S^c) \varpi(dx).$$

*Then*

$$\text{Gap}_R(P) \geq \left( 2 \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right)^{-1}.$$



The fractional lemma gives the following convenient harmonic-sum formulation, which we use to combine the flow bounds in [Proposition 3.4](#).

**Theorem 3.6.** *Let  $P, K_1, \dots, K_N$  be Markov kernels that are reversible with respect to a distribution  $\varpi$ . Suppose*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f), \quad \gamma_j > 0,$$

for every  $f \in L^2(\varpi)$ .

Suppose further that there exist constants  $\phi_1, \dots, \phi_N > 0$  such that every measurable set  $S$  with  $0 < \varpi(S) \leq 1/2$  can be assigned an index  $j(S) \in \{1, \dots, N\}$  satisfying

$$J_{K_{j(S)}}(S, S^c) \geq \phi_{j(S)} \varpi(S).$$

Then

$$\text{Gap}_R(P) \geq \left( 2 \sum_{j=1}^N \frac{1}{\gamma_j \phi_j^2} \right)^{-1}.$$

*Proof.* Apply [Lemma 3.5](#) with

$$\beta_j = \frac{1}{\phi_j}, \quad j = 1, \dots, N.$$

For every measurable  $S$  with  $0 < \varpi(S) \leq 1/2$ ,

$$\begin{aligned} \sum_{j=1}^N \beta_j J_{K_j}(S, S^c) &\geq \frac{J_{K_{j(S)}}(S, S^c)}{\phi_{j(S)}} \\ &\geq \varpi(S). \end{aligned}$$

The conclusion follows directly from [Lemma 3.5](#). □

## 4 Proof of the Main Theorem

*Proof of Theorem 2.1.* Let  $b_0 \leq 1/2$  and  $A_0 \geq 2$  be given in [Proposition 3.4](#), and recall that  $p_\star = A_0\{1 + \log(d+1) + \log \kappa\}$ . We divide the proof into three parts.

*Part 1: The geometric moment ladder.*

Suppose for now that  $p_\star < d$ . Let

$$p_j = 4^j p_\star, \quad j = 0, 1, \dots, J, \tag{8}$$

where  $J$  is the smallest integer for which  $p_J \geq d$ . Thus

$$d \leq p_J < 4d. \tag{9}$$

Define the nominal endpoints

$$\tau_j = \frac{b_0}{L\sqrt{p_j(d+p_j)}},$$

and put

$$\theta = \min \left\{ 1, \frac{H}{\tau_0} \right\}, \quad t_j = \theta \tau_j. \tag{10}$$

Then  $t_0 \leq H$ , and

$$\frac{t_{j+1}}{t_j} = \frac{1}{2} \sqrt{\frac{d+p_j}{d+4p_j}} < \frac{1}{2}.$$

Consequently, the Borel sets  $[t_j/2, t_j]$  are pairwise disjoint, have Lebesgue measure  $t_j/2$ , and lie within  $[0, H]$ . Applying [Lemma 3.1](#) to this finite family gives

$$\mathcal{E}_{\overline{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f). \quad (11)$$

This allows us to apply [Theorem 3.6](#).

Let  $S$  be measurable with  $0 < \pi(S) \leq 1/2$ . We assign  $S$  to a component  $\mathcal{K}_{t_j}$ . Since  $p_0 = p_\star \geq A_0 \geq 2$ , one has  $p_0/2 \geq 1 > \log 2$ . The intervals  $[\log 2, p_0/2]$ ,  $(p_{j-1}/2, p_j/2]$  for  $1 \leq j \leq J$ , and  $(p_J/2, \infty)$  are disjoint and cover  $[\log 2, \infty)$ . Thus the following assignment is exhaustive and unambiguous.

- If  $\log 2 \leq \log(1/\pi(S)) \leq p_0/2$ , assign  $S$  to  $\mathcal{K}_{t_0}$ . By the first assertion of [Proposition 3.4](#),

$$J_{\mathcal{K}_{t_0}}(S, S^c) \geq \phi_0 \pi(S), \quad \phi_0 = 2^{-13} \sqrt{mt_0 \log 2}.$$

- For  $1 \leq j \leq J$ , if  $p_{j-1}/2 < \log(1/\pi(S)) \leq p_j/2$ , assign  $S$  to  $\mathcal{K}_{t_j}$ . By the first assertion of [Proposition 3.4](#),

$$J_{\mathcal{K}_{t_j}}(S, S^c) \geq \phi_j \pi(S), \quad \phi_j = 2^{-13} \sqrt{\frac{mt_j p_{j-1}}{2}} = 2^{-13} \sqrt{\frac{mt_j p_j}{8}}.$$

- If  $\log(1/\pi(S)) > p_J/2$ , assign  $S$  to the last component  $\mathcal{K}_{t_J}$ . By (9) and the assumption  $b_0 \leq 1/2$ ,

$$t_J \leq \frac{b_0}{L\sqrt{p_J(d+p_J)}} \leq \frac{1}{2Ld}.$$

By the second assertion of [Proposition 3.4](#),

$$J_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi'_J \pi(S), \quad \phi'_J = 2^{-13} \min \left\{ 1, \sqrt{\frac{mt_J p_J}{2}} \right\}.$$

Moreover,

$$\frac{mt_J p_J}{2} \leq \frac{b_0}{2\kappa} \sqrt{\frac{p_J}{d+p_J}} \leq \frac{b_0}{2} < 1.$$

Weakening this bound by a constant factor, we obtain

$$J_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi_J \pi(S), \quad \phi_J = 2^{-13} \sqrt{\frac{mt_J p_J}{8}}.$$

We may therefore apply [Theorem 3.6](#) to (11), with

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-26} mt_0 \log 2, \quad \phi_j^2 = 2^{-29} mt_j p_j \quad (j \geq 1).$$

The summation estimate in [Lemma F.1](#) yields

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{HL^2 p_\star (d+p_\star)}{m\theta^2 b_0^2},$$

where  $C$  is universal. Hence,

$$\text{Gap}_R(\bar{P}_H) \geq \frac{m\theta^2 b_0^2}{2CHL^2 p_*(d+p_*)} = \frac{m}{2CH} \min \left\{ H, \frac{b_0}{L\sqrt{p_*(d+p_*)}} \right\}^2. \quad (12)$$

*Part 2: A conservative bound*

The bound (12) is almost what we need, but it has been established only for  $p_* < d$ . To obtain the exact bound in Theorem 2.1, we derive a lower bound on  $\text{Gap}_R(\bar{P}_H)$  using only conservative step sizes.

Define the safe step size

$$t_s = \min \left\{ H, \frac{b_0}{Ld} \right\}.$$

Because  $b_0 \leq 1/2$ ,  $t_s \leq 1/(2Ld)$ . For every measurable  $S$  with  $0 < \pi(S) \leq 1/2$ , apply the second assertion of Proposition 3.4 to obtain

$$J_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt_s \log(1/\pi(S))} \right\} \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt_s \log 2} \right\}. \quad (13)$$

Since

$$mt_s \log 2 \leq \frac{b_0 \log 2}{\kappa d} < 1,$$

the bound (13) becomes

$$J_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-13} \pi(S) \sqrt{mt_s \log 2}.$$

By Lemma 3.1,

$$\mathcal{E}_{\bar{P}_H}(f, f) \geq \frac{t_s}{2H} \mathcal{E}_{\mathcal{K}_{t_s}}(f, f).$$

Apply Theorem 3.6 with  $N = 1$ ,  $K_1 = \mathcal{K}_{t_s}$ ,  $\gamma_1 = t_s/(2H)$ , and  $\phi_1 = 2^{-13} \sqrt{mt_s \log 2}$  to obtain

$$\text{Gap}_R(\bar{P}_H) \geq 2^{-28} (\log 2) \frac{mt_s^2}{H} = 2^{-28} (\log 2) \frac{m}{H} \min \left\{ H, \frac{b_0}{Ld} \right\}^2. \quad (14)$$

*Part 3: Completion of the proof*

If  $p_* \geq d$ , then

$$\frac{1}{\sqrt{p_*(d+p_*)}} \leq \frac{1}{d},$$

so the safe bound (14) already gives (2). If  $p_* < d$ , combine (12) and (14). For positive  $a, b, H$ ,

$$\max\{\min(H, a)^2, \min(H, b)^2\} = \min\{H, \max(a, b)\}^2.$$

Then (2) follows.

It remains to verify the assertion about lazification. Since the identity kernel has zero Dirichlet form,

$$\mathcal{E}_{(I+\bar{P}_H)/2}(f, f) = \frac{1}{2} \mathcal{E}_{\bar{P}_H}(f, f),$$

so

$$\text{Gap}_R((I + \bar{P}_H)/2) = \frac{1}{2} \text{Gap}_R(\bar{P}_H).$$

□

## 5 Lean verification

An accompanying Lean 4 development formally verifies the principal theorem-level results of this article. In particular, it proves [Theorem 2.1](#) for both the non-lazy randomized MALA kernel and its concrete half-lazification, both conclusions of [Corollary 2.2](#), the fixed-step minimax upper bound in [Proposition 2.3](#), the full parameter ranges in [Proposition 3.4](#), and the fractional and hard-assignment aggregation results in [Lemma 3.5](#) and [Theorem 3.6](#). The local-overlap and isoperimetric inputs in [Propositions 3.2](#) and [3.3](#), the geometric step-size ladder, and the explicit universal-constant estimates are also proved within the development rather than introduced as assumptions.

The formal version of the main theorem begins with the same  $C^2$  Hessian hypotheses as the paper. Its Hessian is the actual second Fréchet derivative of  $U$ . A general bridge theorem derives the lower and upper Taylor inequalities and the  $L$ -Lipschitz property of the gradient from the quadratic-form Hessian bounds. The development then constructs the normalized target measure and the fixed- and randomized-step MALA kernels and proves their measurability and reversibility. It represents the paper’s spectral gap by the extended-valued infimum of the Rayleigh quotient over measurable  $L^2$  functions with nonzero variance and proves its equivalence to the corresponding  $L^2$  Poincaré formulation. The zero-variance and infinite-energy cases are included. It also proves, for a general Markov kernel, that half-lazification exactly halves both Dirichlet energy and the Rayleigh spectral gap.

The proof of [Lemma 3.5](#) has precisely the stated  $L^2$  scope. It treats zero component weights explicitly and uses bounded truncations and monotone convergence when applying the energy-domination assumption. The hard-assignment result [Theorem 3.6](#) is then obtained by taking  $\beta_j = \phi_j^{-1}$ . The formalization of [Proposition 2.3](#) verifies the displayed cosine-perturbed potential directly: it proves  $C^\infty$  regularity and the Hessian bounds, identifies the Gaussian first marginal, establishes the local test-function and sticky-set estimates, and carries out the final optimization over the step size and the potential class. Thus the fixed-step proposition does not rely on an assumed hard-target or concentration certificate.

The development does not reproduce every intermediate argument in Appendix B line by line. In particular, it does not formalize the continuous-time Davies and change-of-path-law route as stated there. Instead, it proves the same stationary-rejection and local-overlap conclusions by a finite Gaussian product-likelihood calculation, a finite Euler/random-walk-Metropolis coupling, and weak-limit stability. The required target enlargement inequality is obtained by a formalized Ornstein–Uhlenbeck interpolation for mollified distance ramps followed by a finite-Euler approximation. These discrete arguments are sufficient for all theorem-level conclusions used in the paper.

Several results were stated at a level intended for reuse beyond MALA. They include the Hessian-to-first-order calculus bridge; equivalence of Rayleigh and Poincaré spectral-gap formulations; measurable parameter mixtures of Markov kernels with exact reversibility and energy identities; exact energy scaling under lazification; fractional finite-component aggregation for general reversible kernels; spectral-gap upper bounds from  $L^2$  test functions and indicator cuts; weak-limit stability of metric-enlargement inequalities; finite-dimensional Gaussian enlargement; and weak-limit closure of centered  $L^p$  density bounds when the reference measures vary. The accompanying package was checked with Lean 4.33.0 and mathlib 4.33.0, without proof placeholders or problem-specific axioms. Its public endpoints are collected in `UniformRandomMALA.AllResults`, and the reusable statements are catalogued in `REUSABLE_RESULTS.md`.

## 6 Numerical experiments

We compare fixed-step MALA and uniformly randomized-step MALA on the perturbed-Gaussian target used in the fixed-step obstruction. To distinguish the oscillation scale built into the target from an algorithmic step size, we denote the former by  $h_0$ :

$$U_{d,h_0}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[ \frac{L+m}{4}x_i^2 - \frac{(L-m)h_0}{2} \cos\left(\frac{x_i}{\sqrt{h_0}}\right) \right]. \quad (15)$$

This is the potential in (18) with  $h = h_0$ ; in particular,  $mI_d \preceq \nabla^2 U_{d,h_0}(x) \preceq LI_d$  for every  $x$ . Throughout,  $m = 0.1$  and  $L = 1$ , so  $\kappa = 10$ .

For an endpoint  $H > 0$ , fixed-step MALA uses  $h = H$  at every iteration, whereas randomized-step MALA draws a fresh  $h \sim \text{Unif}(0, H)$  at every iteration. We report the mean acceptance probability and the expected squared jumping distance per coordinate,

$$\text{ESJD} = \frac{1}{d} \mathbb{E} \left[ \alpha_h(X, Y) \|Y - X\|^2 \right]. \quad (16)$$

Both estimates average the conditional acceptance probability  $\alpha_h(X, Y)$  rather than generating the final accept-reject uniform variable. More precisely, for independent one-step experiments  $(X_i, h_i, Y_i)_{i=1}^N$ , the Rao-Blackwellized contributions are

$$R_i^{\text{acc}} = \alpha_{h_i}(X_i, Y_i), \quad R_i^{\text{ESJD}} = \frac{\alpha_{h_i}(X_i, Y_i)}{d} \|Y_i - X_i\|^2.$$

For either contribution  $R_i$ , we report  $\bar{R}_N = N^{-1} \sum_{i=1}^N R_i$  and estimate its standard error directly from the individual observations by

$$\widehat{\text{se}}(\bar{R}_N) = \left\{ \frac{1}{N(N-1)} \sum_{i=1}^N (R_i - \bar{R}_N)^2 \right\}^{1/2}. \quad (17)$$

Thus no batching enters either the point estimates or their standard errors. Rao-Blackwellization reduces Monte Carlo noise without changing the target expectations. All error bars extend two estimated standard errors on either side of the estimate.

### 6.1 Acceptance at the origin

For each  $d \in \{25, 50, 100, 200, 400\}$ , we set

$$h_0 = \frac{4}{L\sqrt{d}}$$

and compare fixed-step MALA with  $h = h_0$  against randomized-step MALA with  $H = h_0$ . Both kernels are evaluated at  $x = 0$ , the point used in the proof of the fixed-step upper bound. Each estimate uses 80,000 independent proposals, and its standard error is computed from (17).

As Figure 1 shows, the fixed-step acceptance probability decreases from approximately  $4.92 \times 10^{-3}$  at  $d = 25$  to  $1.94 \times 10^{-6}$  at  $d = 400$ . The randomized-step acceptance probability decreases much more slowly, from 0.304 to 0.190. Because rejection leaves a chain initialized at the origin unchanged, the time to its first accepted move is geometric. At  $d = 400$ , the reciprocal acceptance estimates imply mean waiting times of approximately  $5.14 \times 10^5$  iterations for fixed-step MALA and 5.25 iterations for randomized-step MALA. The experiment therefore directly displays the sticky behavior

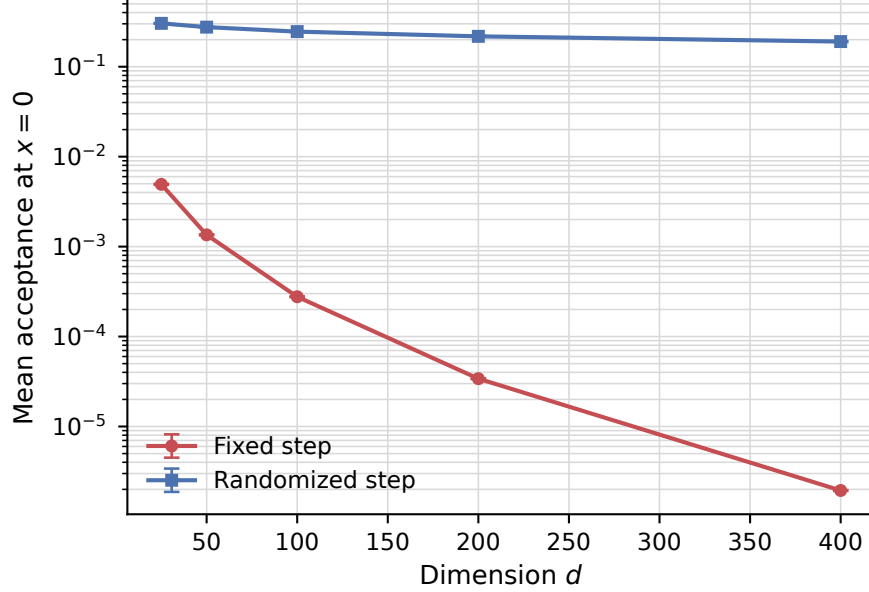


Figure 1: Estimated mean acceptance probability at the origin for the target (15). Fixed-step MALA uses  $h = h_0$ , while randomized-step MALA uses  $h \sim \text{Unif}(0, h_0)$ . The vertical axis is logarithmic, and vertical bars show  $\pm 2$  i.i.d.-estimated standard errors; most are narrower than the markers.

behind the fixed-step obstruction: proposals at the oscillation scale  $h_0$  have rapidly deteriorating acceptance, whereas uniform randomization retains a substantial probability of selecting a smaller step.

To continue the origin experiment beyond one step, we next examine what happens after fixed-step MALA eventually escapes the hard point. We run one chain of each type for 40,000 iterations on the  $d = 200$  target, starting both chains at the origin and taking  $H = h_0$ . The fixed chain uses  $h = h_0$  throughout, whereas the randomized chain independently redraws  $h \sim \text{Unif}(0, h_0)$  at every iteration. We record the first coordinate at every iteration; no thinning is used.

In this realization, the fixed-step chain rejects its first 29,438 proposals and first moves at iteration 29,439. It then makes 5,882 accepted moves by iteration 40,000, so its overall acceptance rate is 0.147. Over the 10,562-iteration interval from its first accepted move through the end of the run, its empirical acceptance rate is 0.557. The trace therefore shows that, once the fixed chain escapes the sticky point, it begins moving on an ordinary scale; the low overall acceptance rate is dominated by the long initial wait. The randomized chain first moves at iteration 2, makes 32,253 accepted moves, and has overall acceptance rate 0.806.

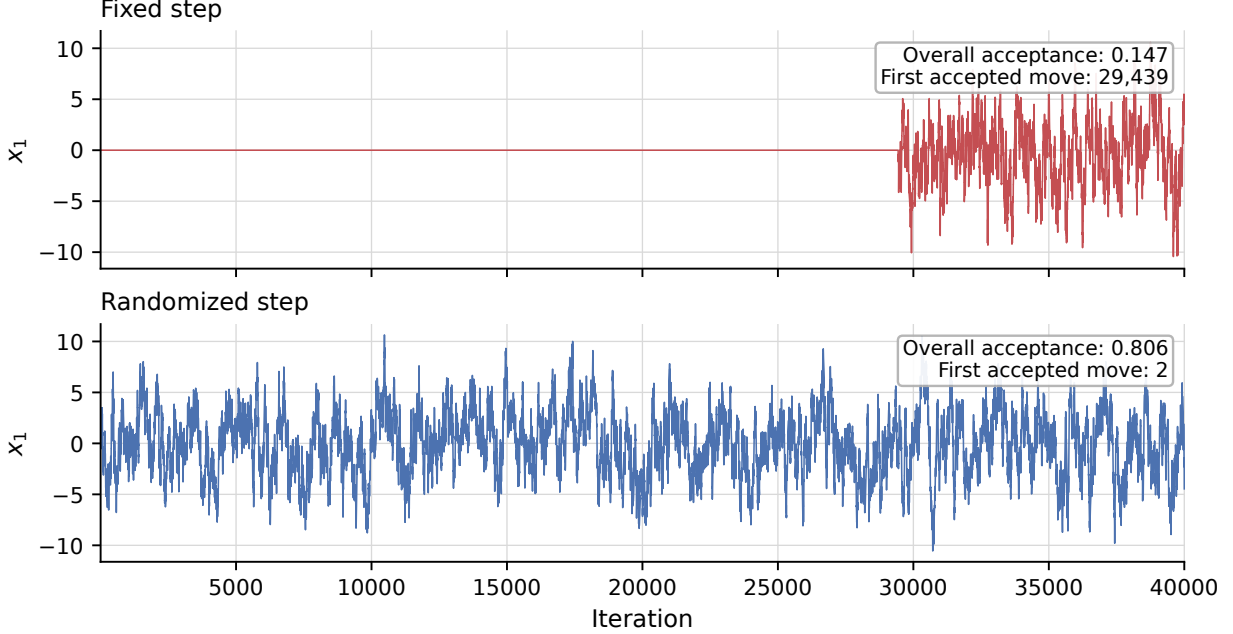


Figure 2: First-coordinate traces continuing the origin experiment for the  $d = 200$  hard target, with both chains initialized at the origin and  $H = h_0 = 4/\sqrt{200}$ . The two panels use common horizontal and vertical scales.

## 6.2 Stationary acceptance and jumping distance

The preceding experiment deliberately starts at an atypical point. We next compare average one-step behavior under stationarity. We fix  $d = 200$  and  $h_0 = 4/\sqrt{200}$ , and vary  $H/h_0$  from 0.05 to 100, using a fine grid near the efficient range and a logarithmically spaced grid on the over-tuning side.

Exact stationary samples are available because (15) is a product target. The first coordinate is  $N(0, m^{-1})$ . For each remaining coordinate, we propose  $\tilde{X} \sim N(0, 2/(L + m))$  and accept it with probability

$$\exp \left\{ \frac{(L - m)h_0}{2} \left[ \cos \left( \frac{\tilde{X}}{\sqrt{h_0}} \right) - 1 \right] \right\}.$$

This is an exact rejection sampler. At every endpoint and for each method, we use 48,000 independent stationary one-step experiments. The standard errors are again computed from the individual contributions using (17).

Figure 3 shows the expected efficiency–robustness tradeoff. At  $H = h_0$ , fixed-step MALA has acceptance 0.559 and ESJD 0.340, while randomized-step MALA has acceptance 0.804 and ESJD 0.216. The largest observed fixed-step ESJD is 0.343, at  $H/h_0 \approx 1.21$ , whereas the largest observed randomized-step ESJD is 0.263, at  $H/h_0 \approx 1.90$ . Thus a well-tuned fixed step has the larger peak ESJD.

Randomization becomes advantageous when the endpoint is too large. At  $H = 2h_0$ , the fixed and randomized ESJDs are 0.230 and 0.261, respectively. At  $H = 3h_0$ , fixed-step acceptance has fallen to 0.0316 and its ESJD to 0.0652, while randomized-step MALA retains acceptance 0.415 and ESJD 0.218. At  $H = 10h_0$ , fixed-step acceptance is approximately  $7.49 \times 10^{-46}$ , whereas randomized-step MALA retains acceptance 0.125 and ESJD 0.0692. Even at  $H = 100h_0$ , the randomized method has acceptance 0.0127 and ESJD 0.00681, while the corresponding fixed-step estimates are below



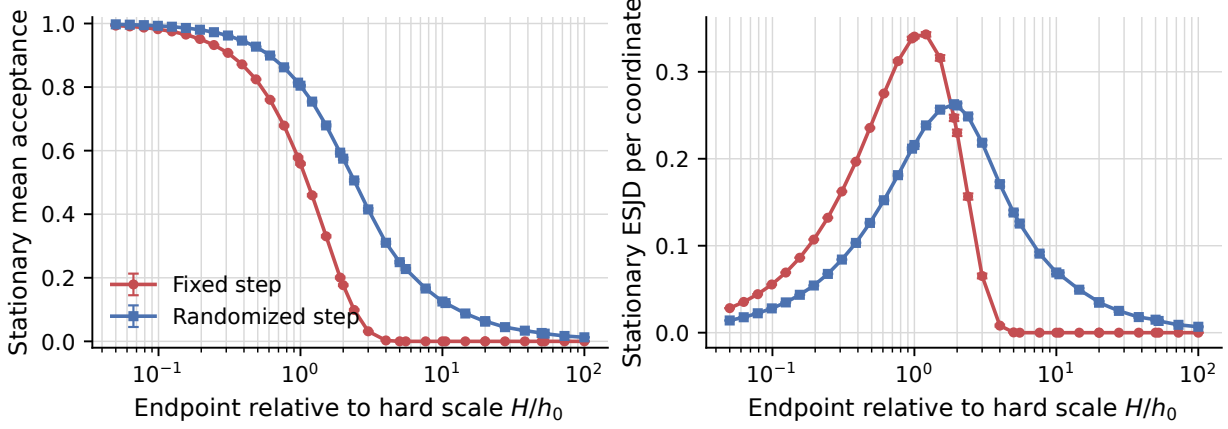


Figure 3: Stationary one-step performance for the  $d = 200$  target with  $h_0 = 4/\sqrt{200}$ . Fixed-step MALA uses  $h = H$ , while randomized-step MALA uses  $h \sim \text{Unif}(0, H)$ . Vertical bars show  $\pm 2$  i.i.d.-estimated standard errors.

floating-point resolution. Randomization does not prevent performance from eventually decreasing as  $H$  grows, but it changes the abrupt fixed-step collapse into a much more gradual decline.

The origin and stationary experiments also expose a distinction relevant to the global spectral gap. When  $d = 200$  and  $H = h_0$ , fixed-step acceptance is about 0.559 under stationarity but only  $3.40 \times 10^{-5}$  at the origin. Thus stationary-average diagnostics largely miss the small sticky region used by the global Poincaré upper bound. Uniform randomization improves acceptance both in that region and under over-tuning, although it does not dominate a well-tuned fixed step in stationary ESJD. These one-step experiments and the single-run trace illustrate tuning robustness and transient behavior; they do not estimate the spectral gap itself.

## A Proof of Proposition 2.3

We follow an existing line of research concerning the mixing properties of fixed-step MALA for hard target distributions. A perturbed-Gaussian target constructed in [Chewi et al., 2021, Theorems 8, 9, and 36] forms the basis of our analysis. Subsequently, Wu et al. [2022, Equation (28) and Lemma 8] combined a perturbed Gaussian block with a Gaussian coordinate, refining the construction. The two works do not readily yield the spectral gap upper bound in Proposition 2.3. Below, we use a modified version of the construction in Wu et al. [2022] to derive the upper bound.

**Proposition A.1.** *There are universal constants  $c, C > 0$  such that the following holds. Let  $d \geq 2$ ,  $0 < m < L$ , and  $h > 0$ . Define*

$$U_{d,h}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[ \frac{L+m}{4}x_i^2 - \frac{(L-m)h}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right) \right]. \quad (18)$$

Then  $U_{d,h}$  is infinitely differentiable and

$$mI_d \preceq \nabla^2 U_{d,h}(x) \preceq LI_d, \quad x \in \mathbb{R}^d. \quad (19)$$

If  $P_h$  is the MALA kernel with step size  $h$  and target density proportional to  $e^{-U_{d,h}}$ , then

$$\text{Gap}_R(P_h) \leq C \min \left\{ mh + \frac{(mh)^2}{2}, \exp[-c(d-1) \min\{(L-m)h, 1\}] \right\}. \quad (20)$$

Writing  $\kappa = L/m$ , this further implies, after possibly changing the universal constants  $c$  and  $C$ , that

$$\text{Gap}_R(P_h) \leq C \min \left\{ \kappa^{-1}Lh, \exp[-cd \min\{(1-\kappa^{-1})Lh, 1\}] \right\}. \quad (21)$$

*Proof.* For  $i \geq 2$ , the  $i$ th diagonal entry of the Hessian of  $U_{d,h}$  is

$$\frac{L+m}{2} + \frac{L-m}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right),$$

which lies in  $[m, L]$ . The first diagonal entry equals  $m$ , and all off-diagonal entries vanish. This proves (19).

We first establish the local upper bound. The target is a product measure, and its first marginal is  $N(0, m^{-1})$ . Use  $f(x) = x_1$  in the Poincaré quotient. Let  $X$  have the target distribution, let  $Z \sim N_d(0, I_d)$  be independent of  $X$ , and write

$$\tilde{Y} = X - h\nabla U_{d,h}(X) + \sqrt{2h}Z$$

for the MALA proposal. The actual transition either moves to  $\tilde{Y}$  or remains at  $X$ . Consequently,

$$\begin{aligned} \mathcal{E}_{P_h}(f, f) &\leq \frac{1}{2} \mathbb{E} \left[ (-hmX_1 + \sqrt{2h}Z_1)^2 \right] \\ &= \frac{1}{2} (h^2m + 2h), \end{aligned}$$

where we used  $\mathbb{E}(X_1^2) = m^{-1}$  and the independence of  $X_1$  and  $Z_1$ . Since  $\text{Var}_\pi(f) = m^{-1}$ , it follows that

$$\text{Gap}_R(P_h) \leq mh + \frac{(mh)^2}{2}. \quad (22)$$

We next obtain the exponential upper bound by controlling the mean acceptance probability at the origin. Write

$$a = \frac{L+m}{2}, \quad b = \frac{L-m}{2}.$$

At the origin,  $\nabla U_{d,h}(0) = 0$ , so a proposal has the form

$$Y = \sqrt{2h} Z.$$

For a general differentiable potential  $U$ , the log Hastings ratio for a proposal from  $x$  to  $y$  is

$$\begin{aligned} R_h(x, y) &:= \log \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)} \\ &= U(x) - U(y) + \frac{1}{2} \langle y - x, \nabla U(x) + \nabla U(y) \rangle \\ &\quad + \frac{h}{4} \left[ \|\nabla U(x)\|^2 - \|\nabla U(y)\|^2 \right]. \end{aligned} \tag{23}$$

For  $i = 2, \dots, d$ , put

$$V_i = \frac{Y_i}{\sqrt{h}} = \sqrt{2} Z_i.$$

Substituting (18) into (23) gives

$$\begin{aligned} R_h(0, Y) &= -\frac{m^2 h^2}{2} Z_1^2 + bh \sum_{i=2}^d A(V_i) \\ &\quad - \frac{h^2}{4} \sum_{i=2}^d [aV_i + b \sin(V_i)]^2, \end{aligned} \tag{24}$$

where

$$A(v) = \cos v - 1 + \frac{v \sin v}{2}.$$

Both the first and the last terms in (24) are nonpositive.

Since  $V_i \sim N(0, 2)$ ,

$$\mathbb{E}(\cos V_i) = e^{-1}, \quad \mathbb{E}(V_i \sin V_i) = 2e^{-1}.$$

The first identity follows from the characteristic function of  $V_i$ , and the second follows from Stein's identity. Hence

$$\mathbb{E}[A(V_i)] = \frac{2}{e} - 1 = -a_0, \quad a_0 = 1 - \frac{2}{e} > 0.$$

Moreover,

$$|A(v)| \leq 2 + \frac{|v|}{2}.$$

Thus  $A(V_i) - \mathbb{E}[A(V_i)]$  has a universally bounded sub-Gaussian norm; see, for example, [Vershynin, 2018, Section 2.5.2]. The concentration inequality for sums of independent sub-Gaussian random variables [Vershynin, 2018, Theorem 2.6.3] therefore yields a universal constant  $c_0 > 0$  such that

$$\mathbb{P} \left( \sum_{i=2}^d A(V_i) > -\frac{a_0(d-1)}{2} \right) \leq e^{-c_0(d-1)}. \tag{25}$$

Outside the event in (25),

$$R_h(0, Y) \leq -\frac{a_0}{2}bh(d-1) = -\frac{a_0}{4}(d-1)(L-m)h.$$

Let

$$a_h(x) = \int \alpha_h(x, y)Q_h(x, dy)$$

denote the pointwise acceptance probability. It follows that

$$\begin{aligned} a_h(0) &= \mathbb{E} \left[ 1 \wedge e^{R_h(0, Y)} \right] \\ &\leq \exp \left[ -\frac{a_0}{4}(d-1)(L-m)h \right] + e^{-c_0(d-1)} \\ &\leq 2 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \end{aligned}$$

for a universal constant  $c_1 > 0$ .

The function  $x \mapsto a_h(x)$  is continuous. Indeed, writing the proposal as

$$x - h\nabla U_{d,h}(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d),$$

the resulting acceptance probability is continuous in  $x$  for every fixed  $Z$  and is bounded by one. Continuity therefore follows from dominated convergence.

Consequently, there is an open ball  $S$  centered at the origin such that

$$0 < \pi(S) \leq \frac{1}{2}, \quad \sup_{x \in S} a_h(x) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}].$$

A transition starting in  $S$  can leave  $S$  only if its proposal is accepted. Hence

$$J_{P_h}(S, S^c) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \pi(S).$$

Testing the Poincaré quotient with  $\mathbf{1}_S$  gives

$$\begin{aligned} \text{Gap}_R(P_h) &\leq \frac{\mathcal{E}_{P_h}(\mathbf{1}_S, \mathbf{1}_S)}{\text{Var}_\pi(\mathbf{1}_S)} \\ &= \frac{J_{P_h}(S, S^c)}{\pi(S)[1 - \pi(S)]} \\ &\leq 8 \exp [-c_1(d-1) \min\{(L-m)h, 1\}]. \end{aligned} \tag{26}$$

Combining (22) and (26) proves (20).

Finally, we prove (21). We have

$$L - m = (1 - \kappa^{-1})L, \quad d - 1 \geq \frac{d}{2}.$$

Therefore

$$(d-1) \min\{(L-m)h, 1\} \geq \frac{d}{2} \min\{(1 - \kappa^{-1})Lh, 1\}.$$

If  $mh \leq 1$ , then

$$mh + \frac{(mh)^2}{2} \leq \frac{3}{2}mh = \frac{3}{2}\kappa^{-1}Lh.$$

If  $mh > 1$ , then

$$\exp [-cd \min\{(1 - \kappa^{-1})Lh, 1\}] \leq 1 < mh = \kappa^{-1}Lh,$$

so the exponential bound is the smaller of the two terms in (21). After adjusting the universal constants, this proves (21).  $\square$

We can now prove [Proposition 2.3](#).

*Proof.* By [Proposition A.1](#), there are universal constants  $c_0, C_0 > 0$  such that, for every  $h > 0$ ,

$$\inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \preceq \nabla^2 U(x) \preceq LI_d, x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq F(Lh), \quad (27)$$

where

$$F(t) = C_0 \min \left\{ \kappa^{-1}t, \exp \left[ -c_0 d \min \{ (1 - \kappa^{-1})t, 1 \} \right] \right\}.$$

Set

$$t_\star = \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})d}.$$

If  $0 < t \leq t_\star$ , then

$$F(t) \leq C_0 \kappa^{-1} t_\star = \frac{2C_0 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}.$$

If  $t > t_\star$ , then the exponential term defining  $F$  is nonincreasing in  $t$ , and hence

$$\begin{aligned} F(t) &\leq C_0 \exp \left[ -c_0 d \min \{ (1 - \kappa^{-1})t_\star, 1 \} \right] \\ &= C_0 \max \left\{ \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} \sup_{t>0} F(t) &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\} \\ &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa_0^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned} \quad (28)$$

Because  $d \geq 2$  and  $\kappa > 1$ ,

$$\frac{1}{(\kappa d)^2} \leq C \frac{\log(\kappa d)}{\kappa d}$$

for a universal constant  $C$ . Absorbing the factors depending only on  $\kappa_0$  into  $C$ , and changing the universal exponential constant if necessary, (27) and (28) prove (3).  $\square$

## B Proof of Proposition 3.2

This appendix proves [Proposition 3.2](#). The main input is the following stationary estimate for the pointwise rejection function

$$r_h(x) = 1 - \int \alpha_h(x, y) Q_h(x, dy),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}, \quad q_h(x, y) = (4\pi h)^{-d/2} \exp \left\{ -\frac{\|y - x + h \nabla U(x)\|^2}{4h} \right\}.$$

**Proposition B.1** (Stationary rejection estimate). *There are universal constants  $c, C > 0$  such that, for every real  $p \geq 2$ ,*

$$h \leq \frac{c}{L\sqrt{p(d+p)}} \implies \|r_h\|_{L^p(\pi)} \leq CLh\sqrt{p(d+p)}. \quad (29)$$

The proof compares the exact stationary Langevin path with the path whose drift is frozen at its initial position. The endpoint of the latter path is exactly the MALA proposal. Similar diffusion-comparison ideas appear in [Bou-Rabee and Vanden-Eijnden \[2010\]](#), [Bou-Rabee and Hairer \[2013\]](#), [Chewi et al. \[2021\]](#).

**Attribution within this appendix.** The stationary linear-increment estimate in [Lemma B.3](#) is a direct consequence of the exponential conjugation estimate in [Lemma B.2](#), which is the elementary  $L^2$  calculation at the core of the Davies exponential-perturbation method for symmetric semigroups [[Davies, 1989](#), Section 3.2, especially Lemma 3.2.1 and p. 85]; see also [Ouhabaz \[2005, Chapter 6, especially Section 6.3\]](#). The same estimate also follows from the Lyons–Zheng forward–backward decomposition [[Fukushima et al., 2011](#), Theorem 5.7.1]; see also the original paper of [Lyons and Zheng \[1988\]](#). [Lemma B.4](#) is an elementary consequence of that linear estimate, Gaussian randomization, and Jensen’s inequality; it is not quoted from another source. The Gaussian-randomization step is the identity-matrix specialization of the argument used for sub-Gaussian quadratic forms by [Hsu et al. \[2012, Theorem 1 and Remark 2\]](#). [Lemma B.5](#) is likewise not quoted verbatim from the literature: it combines standard results on stochastic exponentials, Novikov’s criterion, Girsanov’s theorem, stopping of stochastic integrals, Doob’s inequality, and the quantitative BDG inequality, with the problem-specific estimate  $V_h \leq L^2 J_h/2$ . General reverse-Hölder theory for continuous stochastic exponentials is developed by [Kazamaki \[1994\]](#), but the dimension- and moment-explicit calculation needed here is supplied in full. The passage from the path likelihood to the rejection probability uses the standard conditional-likelihood identity and the Metropolis–Hastings meet identity [[Kallenberg, 2021](#), [Tierney, 1998](#), [Billera and Diaconis, 2001](#)]. Thus [Proposition B.1](#) synthesizes these ingredients for the present MALA problem.

## B.1 Stationary Langevin increments

We begin by fixing the continuous-time framework used throughout this appendix. All random variables are defined on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$$

satisfying the usual completeness and right-continuity conditions. A standard Brownian motion relative to this filtration is an adapted continuous process  $B$  with  $B_0 = 0$  such that, for  $0 \leq s < t$ , the increment  $B_t - B_s$  has law  $N_d(0, (t - s)I_d)$  and is independent of  $\mathcal{F}_s$ . This last independence property will be used explicitly in [Appendix B.2](#). Basic references for filtered probability spaces, Brownian motion, Itô integration, and stochastic differential equations are [Karatzas and Shreve \[1991, Chapters 3 and 5\]](#), [Revuz and Yor \[1999, Chapters IV and IX\]](#), and [Kallenberg \[2021\]](#).

Because  $\nabla U$  is globally Lipschitz, for every  $x \in \mathbb{R}^d$  the Langevin equation has a unique nonexplosive strong solution  $(X_t^x)_{t \geq 0}$  with  $X_0^x = x$ ; see, for example, [Karatzas and Shreve \[1991, Chapter 5\]](#). For a bounded measurable function  $f$ , define the transition semigroup pointwise by

$$S_t f(x) = \mathbb{E}[f(X_t^x)].$$

Now let  $X_0$  be  $\mathcal{F}_0$ -measurable and independent of  $B$ , with law  $\pi$ , and let  $(X_t)_{t \geq 0}$  be the corresponding solution of

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad X_0 \sim \pi. \quad (30)$$

The differential notation means the integral identity

$$X_t = X_0 - \int_0^t \nabla U(X_s) ds + \sqrt{2} B_t, \quad t \geq 0. \quad (31)$$

Its generator on  $C_c^\infty(\mathbb{R}^d)$  is

$$\mathcal{L}f = \Delta f - \langle \nabla U, \nabla f \rangle.$$

Integration by parts against  $\pi(dx) \propto e^{-U(x)} dx$  gives, for  $f, g \in C_c^\infty(\mathbb{R}^d)$ ,

$$\int f \mathcal{L}g \, d\pi = - \int \langle \nabla f, \nabla g \rangle \, d\pi = \int g \mathcal{L}f \, d\pi.$$

Let  $W^{1,2}(\pi)$  denote the completion of  $C_c^\infty(\mathbb{R}^d)$  for the norm

$$\|f\|_{W^{1,2}(\pi)}^2 = \int (f^2 + \|\nabla f\|^2) \, d\pi.$$

If  $\text{Dom}(\mathcal{L})$  denotes the domain of the  $L^2(\pi)$  generator, then  $\text{Dom}(\mathcal{L}) \subset W^{1,2}(\pi)$ , and the integration-by-parts identity extends to

$$\langle f, \mathcal{L}w \rangle_{L^2(\pi)} = - \int \langle \nabla f, \nabla w \rangle \, d\pi, \quad f \in W^{1,2}(\pi), \quad w \in \text{Dom}(\mathcal{L}). \quad (32)$$

Standard diffusion-semigroup theory shows that the self-adjoint semigroup generated by  $\mathcal{L}$  solves the same martingale problem as (30); uniqueness in law therefore identifies it with  $(S_t)$ . Thus  $\pi$  is invariant and  $S_t$  is self-adjoint in  $L^2(\pi)$ . Since  $X_0 \sim \pi$ , the process is stationary and reversible. These standard identifications and domain facts are discussed in Fukushima et al. [2011, Chapters 1 and 7]; see also Ouhabaz [2005, Sections 1.3–1.5] and Bakry et al. [2014].

We next record the weighted-semigroup estimate used below.

**Lemma B.2.** *Let  $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$  be bounded and belong to  $W^{1,\infty}$ , and set*

$$T_t^\psi g = e^{-\psi} S_t(e^\psi g).$$

*Then*

$$\|T_t^\psi\|_{L^2(\pi) \rightarrow L^2(\pi)} \leq \exp(t \|\nabla \psi\|_\infty^2), \quad t \geq 0. \quad (33)$$

*Proof.* This is the  $p = 2$  energy calculation at the core of Davies' exponential-perturbation method; see Davies [1989, Section 3.2, especially Lemma 3.2.1 and pp. 83–85]. For the Sobolev-domain and semigroup facts used below, see Ouhabaz [2005, Sections 1.3–1.5 and 6.3].

Fix  $g \in L^2(\pi)$ , and put

$$u_t = S_t(e^\psi g), \quad v_t = e^{-\psi} u_t.$$

Since  $\mathcal{L}$  is self-adjoint and nonpositive, the spectral theorem gives  $\|\mathcal{L}S_t\|_{L^2(\pi) \rightarrow L^2(\pi)} \leq 1/(et)$ . In particular, for every  $t > 0$ ,  $u_t \in \text{Dom}(\mathcal{L})$  and

$$\frac{du_t}{dt} = \mathcal{L}u_t \quad \text{in } L^2(\pi).$$

Multiplication by  $e^{-\psi}$  is bounded on  $L^2(\pi)$ ; hence  $t \mapsto v_t$  is also differentiable for  $t > 0$ , with

$$\frac{dv_t}{dt} = e^{-\psi} \mathcal{L}u_t.$$

For any differentiable curve in  $L^2(\pi)$ , the squared norm is differentiated by

$$\frac{\|v_{t+\varepsilon}\|_{L^2(\pi)}^2 - \|v_t\|_{L^2(\pi)}^2}{\varepsilon}$$

$$= \langle (v_{t+\varepsilon} - v_t)/\varepsilon, v_{t+\varepsilon} + v_t \rangle_{L^2(\pi)} \longrightarrow 2 \langle dv_t/dt, v_t \rangle_{L^2(\pi)}.$$

Therefore

$$\frac{1}{2} \frac{d}{dt} \|v_t\|_{L^2(\pi)}^2 = \int v_t \frac{dv_t}{dt} d\pi.$$

We now apply (32). The inclusion  $u_t \in \text{Dom}(\mathcal{L})$  implies  $u_t \in W^{1,2}(\pi)$ . Moreover, multiplication by a bounded  $W^{1,\infty}$  function preserves  $W^{1,2}(\pi)$ , by the weak product rule. Thus  $v_t = e^{-\psi} u_t$  and  $e^{-\psi} v_t$  belong to  $W^{1,2}(\pi)$ . Since  $u_t = e^{\psi} v_t$ , we obtain

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|v_t\|_{L^2(\pi)}^2 &= \int e^{-\psi} v_t \mathcal{L}(e^{\psi} v_t) d\pi \\ &= - \int \langle \nabla(e^{-\psi} v_t), \nabla(e^{\psi} v_t) \rangle d\pi. \end{aligned}$$

The weak product and chain rules give

$$\nabla(e^{-\psi} v_t) = e^{-\psi} (\nabla v_t - v_t \nabla \psi), \quad \nabla(e^{\psi} v_t) = e^{\psi} (\nabla v_t + v_t \nabla \psi).$$

Consequently, the two cross terms cancel almost everywhere:

$$\begin{aligned} \langle \nabla(e^{-\psi} v_t), \nabla(e^{\psi} v_t) \rangle &= \langle \nabla v_t - v_t \nabla \psi, \nabla v_t + v_t \nabla \psi \rangle \\ &= \|\nabla v_t\|^2 - v_t^2 \|\nabla \psi\|^2. \end{aligned}$$

It follows that

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \|v_t\|_{L^2(\pi)}^2 &= - \int \|\nabla v_t\|^2 d\pi + \int v_t^2 \|\nabla \psi\|^2 d\pi \\ &\leq \|\nabla \psi\|_{\infty}^2 \|v_t\|_{L^2(\pi)}^2. \end{aligned}$$

Gronwall's inequality, applied between  $s$  and  $t$  with  $0 < s < t$ , yields

$$\|v_t\|_{L^2(\pi)}^2 \leq \exp(2(t-s) \|\nabla \psi\|_{\infty}^2) \|v_s\|_{L^2(\pi)}^2.$$

Finally, the strong continuity of  $S_t$  gives

$$\|v_s - g\|_{L^2(\pi)} \leq \|e^{-\psi}\|_{\infty} \|S_s(e^{\psi} g) - e^{\psi} g\|_{L^2(\pi)} \longrightarrow 0 \quad (s \downarrow 0).$$

Letting  $s \downarrow 0$ , and then taking square roots, proves (33). The case  $t = 0$  is immediate from  $S_0 = I$ .  $\square$

**Lemma B.3.** *For every  $v \in \mathbb{R}^d$ ,  $\lambda \in \mathbb{R}$ , and  $t \geq 0$ ,*

$$\mathbb{E}[\exp(\lambda \langle v, X_t - X_0 \rangle)] \leq \exp(\lambda^2 t \|v\|^2). \quad (34)$$

*Proof.* Choose smooth functions  $\chi_R : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$\chi_R(s) = s \quad \text{for } |s| \leq R, \quad |\chi_R'(s)| \leq 1, \quad \chi_R \text{ is bounded,}$$

and set

$$\psi_R(x) = \lambda \chi_R(\langle v, x \rangle).$$



Then  $\|\nabla\psi_R\|_\infty \leq |\lambda| \|v\|$ . Because  $X_0 \sim \pi$ , the Markov property and stationarity give

$$\begin{aligned}\mathbb{E}[e^{\psi_R(X_t) - \psi_R(X_0)}] &= \int e^{-\psi_R(x)} S_t(e^{\psi_R})(x) \pi(dx) \\ &= \langle 1, T_t^{\psi_R} 1 \rangle_{L^2(\pi)}.\end{aligned}$$

Since  $\pi$  is a probability measure,  $\|1\|_{L^2(\pi)} = 1$ . Cauchy–Schwarz and (33) therefore imply

$$\mathbb{E}[e^{\psi_R(X_t) - \psi_R(X_0)}] \leq \|T_t^{\psi_R}\|_{2 \rightarrow 2} \leq e^{\lambda^2 t \|v\|^2}.$$

For every sample path,  $\psi_R(X_t) - \psi_R(X_0) \rightarrow \lambda \langle v, X_t - X_0 \rangle$  as  $R \rightarrow \infty$ . The integrands are nonnegative, so Fatou’s lemma gives (34).  $\square$

**Alternative literature proof.** For a symmetric Markov process started from its symmetrizing measure, the Lyons–Zheng decomposition expresses a stationary increment as one half of a forward martingale minus one half of a reverse-time martingale. Applied to  $x \mapsto \langle v, x \rangle$ , each martingale endpoint has deterministic quadratic variation  $2t \|v\|^2$ , and hence is centered Gaussian by Lévy’s characterization. Cauchy–Schwarz then gives exactly (34). See Fukushima et al. [2011, Section 5.7, especially Theorem 5.7.1], the original construction of Lyons and Zheng [1988], and a concise later formulation due to Kuwae and Uemura [1997].

The next proof uses a standard Gaussian-randomization argument. In a more general matrix form, it is the argument used by Hsu et al. [2012, Theorem 1 and Remark 2] to pass from linear sub-Gaussian moment bounds to quadratic-form bounds. We include the identity-matrix calculation because the exact constant in (35) is useful below.

**Lemma B.4.** *For every  $t > 0$  and  $0 \leq a < 1/(4t)$ ,*

$$\mathbb{E}[\exp(a \|X_t - X_0\|^2)] \leq (1 - 4at)^{-d/2}. \quad (35)$$

*If*

$$J_h = \int_0^h \|X_t - X_0\|^2 dt,$$

*then, for  $0 \leq \lambda < 1/(4h^2)$ ,*

$$\mathbb{E}(e^{\lambda J_h}) \leq (1 - 4\lambda h^2)^{-d/2}. \quad (36)$$

*Consequently, for every real  $p \geq 1$ ,*

$$\|J_h\|_{L^p(\mathbb{P})} \leq Ch^2(d + p), \quad (37)$$

*where  $C$  is universal.*

*Proof.* Enlarge the probability space, if necessary, so that it also carries a random vector  $G \sim N_d(0, I_d)$  independent of the entire Langevin path  $(X_s)_{s \geq 0}$ . For a fixed vector  $z \in \mathbb{R}^d$ , the scalar  $\langle G, z \rangle$  is Gaussian with mean zero and variance  $\|z\|^2$ . The Gaussian moment-generating formula therefore gives

$$\mathbb{E}(e^{\sqrt{2a}\langle G, z \rangle}) = e^{a\|z\|^2}.$$

Apply this identity with the random vector  $z = X_t - X_0$ . More precisely, conditioning first on the Langevin path gives

$$\mathbb{E}(e^{a\|X_t - X_0\|^2}) = \mathbb{E}\left[\mathbb{E}\left(e^{\sqrt{2a}\langle G, X_t - X_0 \rangle} \mid \sigma(X_s : s \geq 0)\right)\right].$$

The integrand is nonnegative, so Tonelli's theorem allows us to reverse the order in which the path and  $G$  are averaged. Conditional on  $G$ , the vector  $G$  is fixed, and [Lemma B.3](#), with  $v = G$  and  $\lambda = \sqrt{2a}$ , gives

$$\mathbb{E} \left( e^{\sqrt{2a} \langle G, X_t - X_0 \rangle} \mid G \right) \leq e^{2at \|G\|^2}.$$

Consequently,

$$\begin{aligned} \mathbb{E} \left( e^{a \|X_t - X_0\|^2} \right) &\leq \mathbb{E} \left( e^{2at \|G\|^2} \right) \\ &= (1 - 4at)^{-d/2}, \end{aligned}$$

where the last equality is the moment-generating function of a  $\chi_d^2$  random variable and requires  $4at < 1$ . This proves [\(35\)](#).

We next pass from one increment to its time integral. For each fixed sample path, Jensen's inequality for the convex function  $x \mapsto e^x$ , applied to the uniform probability measure  $dt/h$  on  $[0, h]$ , gives

$$\begin{aligned} e^{\lambda J_h} &= \exp \left( \frac{1}{h} \int_0^h \lambda h \|X_t - X_0\|^2 dt \right) \\ &\leq \frac{1}{h} \int_0^h e^{\lambda h \|X_t - X_0\|^2} dt. \end{aligned}$$

Take expectations and use Tonelli's theorem, followed by [\(35\)](#) with  $a = \lambda h$ :

$$\begin{aligned} \mathbb{E}(e^{\lambda J_h}) &\leq \frac{1}{h} \int_0^h \mathbb{E} e^{\lambda h \|X_t - X_0\|^2} dt \\ &\leq \frac{1}{h} \int_0^h (1 - 4\lambda h t)^{-d/2} dt \\ &\leq (1 - 4\lambda h^2)^{-d/2}. \end{aligned}$$

This proves [\(36\)](#).

It remains to derive the  $L^p$  estimate from this exponential-tail bound. Put  $\lambda = (8h^2)^{-1}$  in [\(36\)](#). Markov's inequality gives, for every  $s \geq 0$ ,

$$\mathbb{P} \left( J_h \geq 4dh^2 \log 2 + 8h^2 s \right) \leq e^{-s}.$$

Define

$$Y = \frac{(J_h - 4dh^2 \log 2)_+}{8h^2}.$$

Then  $\mathbb{P}(Y > s) \leq e^{-s}$ . For every real  $p \geq 1$ , the layer-cake formula for moments gives

$$\begin{aligned} \mathbb{E}(Y^p) &= p \int_0^\infty s^{p-1} \mathbb{P}(Y > s) ds \\ &\leq p \int_0^\infty s^{p-1} e^{-s} ds = \Gamma(p+1). \end{aligned}$$

Stirling's bound implies  $\Gamma(p+1)^{1/p} \leq Cp$  for all  $p \geq 1$ . Hence

$$\|J_h\|_{L^p(\mathbb{P})} \leq 4dh^2 \log 2 + 8h^2 \|Y\|_{L^p(\mathbb{P})} \leq Ch^2(d+p),$$

which proves [\(37\)](#). For the tail-integration identity and the Gamma-function estimate, see [Boucheron et al. \[2013, Chapter 2\]](#).  $\square$

## B.2 Girsanov comparison with frozen drift

We compare two laws on the same path space over  $[0, h]$  and assume throughout this subsection that  $Lh < 1$ . This entails no loss in the final theorem, because the step-size condition in [Proposition B.1](#) implies  $Lh < 1$  once its universal constant is chosen sufficiently small. Under the original probability measure  $\mathbb{P}$ , the path satisfies the exact Langevin equation (30). We shall construct a second measure under which the coordinate process has drift frozen at the initial value  $-\nabla U(X_0)$ .

We recall the continuous-martingale facts used below. If  $H_t$  is a predictable  $\mathbb{R}^d$ -valued process with  $\int_0^h \|H_t\|^2 dt < \infty$  almost surely, then  $\int_0^t \langle H_s, dB_s \rangle$  is a continuous local martingale. Its quadratic variation is

$$\left[ \int_0^\cdot \langle H_s, dB_s \rangle \right]_t = \int_0^t \|H_s\|^2 ds.$$

Quadratic variation is the continuous-time analogue of the accumulated conditional variance of a discrete-time martingale sum. For a continuous local martingale  $N$  with  $N_0 = 0$ , the stochastic exponential is

$$\mathcal{E}(N)_t = \exp\{N_t - [N]_t/2\}.$$

It is always a nonnegative local martingale, but it need not have expectation one. Novikov's criterion states that  $\mathbb{E} \exp\{[N]_h/2\} < \infty$  is sufficient for  $(\mathcal{E}(N)_t)_{0 \leq t \leq h}$  to be a true mean-one martingale. These statements, together with Itô's formula and the stopping arguments used below, are standard; see [Karatzas and Shreve \[1991, Chapter 3\]](#), [Revuz and Yor \[1999, Chapters IV and VIII\]](#), or [Kallenberg \[2021\]](#).

Under the exact stationary law, define

$$\theta_t = \frac{\nabla U(X_t) - \nabla U(X_0)}{\sqrt{2}}, \quad V_t = \int_0^t \|\theta_s\|^2 ds, \quad M_t = \int_0^t \langle \theta_s, dB_s \rangle.$$

The process  $\theta$  is continuous and adapted, hence predictable. Since  $X$  has continuous paths,  $\int_0^h \|\theta_s\|^2 ds < \infty$  almost surely on every finite interval, so the Itô integral is well defined as a continuous local martingale. Its quadratic variation is  $[M]_t = V_t$ . The Lipschitz property of  $\nabla U$  and [Lemma B.4](#) give

$$V_h \leq \frac{L^2}{2} J_h, \quad \|V_h\|_{L^p(\mathbb{P})} \leq CL^2 h^2 (d + p), \quad p \geq 1. \quad (38)$$

Define

$$D_t = \exp\left(M_t - \frac{1}{2} V_t\right) = \mathcal{E}(M)_t.$$

Before using  $D_h$  as a likelihood ratio, we must verify that it has mean one. If  $Lh < 1$ , then (36) and (38) imply

$$\begin{aligned} \mathbb{E}(e^{[M]_h/2}) &= \mathbb{E}(e^{V_h/2}) \\ &\leq \mathbb{E}(e^{L^2 J_h/4}) \leq (1 - L^2 h^2)^{-d/2} < \infty. \end{aligned}$$

Novikov's criterion therefore shows that  $(D_t)_{0 \leq t \leq h}$  is a true  $\mathbb{P}$ -martingale and

$$\mathbb{E}(D_t \mid \mathcal{F}_s) = D_s, \quad 0 \leq s \leq t \leq h. \quad (39)$$

We may now define a probability measure  $\mathbb{P}_h^{\text{fr}}$  on  $(\Omega, \mathcal{F}_h)$  by

$$\frac{d\mathbb{P}_h^{\text{fr}}}{d\mathbb{P}} \Big|_{\mathcal{F}_h} = D_h.$$

The superscript “fr” indicates frozen drift. Girsanov’s theorem states that, under  $\mathbb{P}_h^{\text{fr}}$ ,

$$W_t = B_t - \int_0^t \theta_s \, ds, \quad 0 \leq t \leq h,$$

is a Brownian motion relative to the same filtration  $(\mathcal{F}_t)_{0 \leq t \leq h}$ ; see [Karatzas and Shreve \[1991, Section 3.5\]](#) or [Revuz and Yor \[1999, Chapter VIII, Section 1\]](#). Substituting  $B_t = W_t + \int_0^t \theta_s \, ds$  into the integral equation (31) gives, under the new measure,

$$\begin{aligned} X_t &= X_0 - \int_0^t \nabla U(X_s) \, ds + \sqrt{2} W_t + \sqrt{2} \int_0^t \theta_s \, ds \\ &= X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \quad 0 \leq t \leq h. \end{aligned}$$

Thus

$$X_t = X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \quad 0 \leq t \leq h, \quad (40)$$

under  $\mathbb{P}_h^{\text{fr}}$ .

We record two conditional facts. First, the change of measure does not alter the law of  $X_0$ . If  $A \subseteq \mathbb{R}^d$  is Borel, then

$$\begin{aligned} \mathbb{P}_h^{\text{fr}}(X_0 \in A) &= \mathbb{E}(D_h \mathbf{1}_{\{X_0 \in A\}}) \\ &= \mathbb{E} \left[ \mathbb{E}(D_h \mid \mathcal{F}_0) \mathbf{1}_{\{X_0 \in A\}} \right] \\ &= \mathbb{P}(X_0 \in A) = \pi(A), \end{aligned}$$

where (39) gives  $\mathbb{E}(D_h \mid \mathcal{F}_0) = D_0 = 1$ .

Second,  $W_h$  is independent of  $\mathcal{F}_0$  under  $\mathbb{P}_h^{\text{fr}}$ . Indeed, Girsanov’s theorem implies that  $W$  is Brownian with respect to  $(\mathcal{F}_t)$ , so  $W_h - W_0 = W_h$  is independent of  $\mathcal{F}_0$  and has law  $N_d(0, hI_d)$ . Since  $X_0$  is  $\mathcal{F}_0$ -measurable,  $W_h$  is also independent of  $X_0$ . Equation (40) therefore shows that, conditional on  $X_0 = x$ ,

$$X_h = x - h \nabla U(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d),$$

so the endpoint distribution under the frozen law is exactly  $Q_h(x, \cdot)$ .

We next quantify how close the path likelihood  $D_h$  is to one. General reverse-Hölder and moment estimates for continuous stochastic exponentials form a classical theory; see, for example, [Kazamaki \[1994, Chapters 1 and 3\]](#). We give a direct localized calculation to track explicitly the dependence on  $d, p, L$ , and  $h$  and to exploit the problem-specific bound  $V_h \leq L^2 J_h / 2$ .

**Lemma B.5.** *There are universal constants  $c, C > 0$  such that, for every real  $p \geq 2$ ,*

$$h \leq \frac{c}{L \sqrt{p(d+p)}} \quad (41)$$

*implies*

$$\|D_h\|_{L^{2p}(\mathbb{P})} \leq C, \quad \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh \sqrt{p(d+p)}. \quad (42)$$

*Proof.* Set

$$q = 2p$$

at the outset. Thus  $q \geq 4$ . No separate condition on an auxiliary moment order is being imposed: the stopped-exponential calculation in Step 1 would work for any  $q > 1$ , while the choice  $q = 2p$  is made to match the Hölder and Doob exponents used in Step 3. The first estimate in (42) is an

$L^q$  estimate, and the same exponent  $q = 2p$  will later be used in Hölder's and Doob's inequalities. Choose a universal constant  $c_1 > 0$ , sufficiently small for the estimates below. Since

$$q(d + q) = 2p(d + 2p) \leq 4p(d + p),$$

the hypothesis (41), with  $c \leq c_1/2$ , implies

$$h \leq \frac{c_1}{L\sqrt{q(d + q)}}. \quad (43)$$

We work under this condition for the rest of the proof.

**Stopping conventions.** We use standard process-level stopping identities, not the informal rule of substituting a random time into a statement known only for each fixed deterministic time. If  $N$  is a continuous local martingale and  $\tau$  is a stopping time, then, up to indistinguishability,

$$N_t^\tau := N_{t \wedge \tau}, \quad [N^\tau]_t = [N]_{t \wedge \tau}.$$

Likewise, for a predictable integrand  $H$ ,

$$\left( \int_0^\cdot \langle H_s, dB_s \rangle \right)_{t \wedge \tau} = \int_0^t \mathbf{1}_{\{s \leq \tau\}} \langle H_s, dB_s \rangle. \quad (44)$$

In particular, if  $\mathcal{E}(N)_t = \exp\{N_t - [N]_t/2\}$  denotes the stochastic exponential, then

$$\mathcal{E}(N^\tau)_t = \exp\left(N_{t \wedge \tau} - \frac{1}{2}[N]_{t \wedge \tau}\right) = \mathcal{E}(N)_{t \wedge \tau}.$$

These are identities of stopped processes and therefore hold for all  $t$  simultaneously outside a single null set; continuity then permits evaluation at any bounded random time. Purely algebraic identities among continuous processes may be evaluated at  $t \wedge \tau$  for the same reason, and inequalities such as  $V_{t \wedge \tau} \leq V_t$  follow pathwise from monotonicity. By contrast, an estimate obtained only after taking expectations at each fixed time cannot generally be evaluated at a stopping time without a separate optional-stopping argument. No such step is used here. See [Revuz and Yor \[1999, Chapter IV, Sections 1–2\]](#), [Karatzas and Shreve \[1991, Chapter 3\]](#), or [Protter \[2005, Chapter II\]](#) for stopping of continuous local martingales, stochastic integrals, and quadratic variation.

*Step 1: a uniform  $L^q$  bound for  $D_h$ .* For  $n \geq 1$ , let

$$\tau_n = \inf\{t \in [0, h] : V_t \geq n\}, \quad \inf \emptyset = \infty.$$

Because  $V$  is continuous and adapted,  $\tau_n$  is a stopping time. Set  $M_t^{(n)} = M_{t \wedge \tau_n}$ . By the standard stopping identity for quadratic variation stated above,

$$[M^{(n)}]_t = [M]_{t \wedge \tau_n} = V_{t \wedge \tau_n}.$$

Consequently, the quadratic variation of the continuous local martingale  $2qM^{(n)}$  is

$$[2qM^{(n)}]_t = 4q^2 V_{t \wedge \tau_n} \leq 4q^2 n, \quad 0 \leq t \leq h.$$

Consider its stochastic exponential

$$\Lambda_t^{(q,n)} = \exp(2qM_{t \wedge \tau_n} - 2q^2 V_{t \wedge \tau_n}).$$

Novikov's criterion applies because

$$\mathbb{E} \left[ \exp \left( \frac{1}{2} [2qM^{(n)}]_h \right) \right] \leq e^{2q^2n} < \infty.$$

Hence  $\Lambda^{(q,n)}$  is a true mean-one martingale, not merely a local martingale, and

$$\mathbb{E} [\Lambda_h^{(q,n)}] = 1. \quad (45)$$

The definitions of  $D$  and  $\Lambda^{(q,n)}$  give the exact algebraic identity

$$D_{h \wedge \tau_n}^q = [\Lambda_h^{(q,n)}]^{1/2} \exp[(q^2 - q/2)V_{h \wedge \tau_n}].$$

Cauchy–Schwarz, (45), and  $V_{h \wedge \tau_n} \leq V_h$  yield

$$\begin{aligned} \mathbb{E}(D_{h \wedge \tau_n}^q) &\leq \{\mathbb{E}[\Lambda_h^{(q,n)}]\}^{1/2} \left\{ \mathbb{E}[e^{(2q^2 - q)V_{h \wedge \tau_n}}] \right\}^{1/2} \\ &\leq \left\{ \mathbb{E}[e^{(2q^2 - q)V_h}] \right\}^{1/2}. \end{aligned}$$

Since  $V_h \leq L^2 J_h / 2$ , (36) gives

$$\mathbb{E}(D_{h \wedge \tau_n}^q) \leq (1 - u)^{-d/4}, \quad u = 2(2q^2 - q)L^2 h^2,$$

provided  $u < 1$ .

We now spell out the  $q$ -th-root calculation. From (43),

$$u \leq 4q^2 L^2 h^2 \leq 4c_1^2 \frac{q}{d + q}.$$

Choose  $c_1$  so that  $4c_1^2 \leq 1/2$ . Then  $0 \leq u \leq 1/2$ , and  $\log(1/(1 - u)) \leq 2u$ . Therefore

$$\begin{aligned} \log \|D_{h \wedge \tau_n}\|_{L^q(\mathbb{P})} &= \frac{1}{q} \log \left( \mathbb{E} D_{h \wedge \tau_n}^q \right) \\ &\leq \frac{d}{4q} \log \frac{1}{1 - u} \\ &\leq \frac{du}{2q} \\ &\leq 2dqL^2 h^2 \\ &\leq 2c_1^2 \frac{d}{d + q} \leq 2c_1^2. \end{aligned}$$

Thus

$$\|D_{h \wedge \tau_n}\|_{L^q(\mathbb{P})} \leq e^{2c_1^2} \quad (46)$$

uniformly in  $n$ . Finally,  $V_h < \infty$  almost surely. For every sample path, once  $n > V_h$ , one has  $\tau_n = \infty$  and hence  $D_{h \wedge \tau_n} = D_h$ . Fatou's lemma applied to the  $q$ -th powers in (46) gives

$$\|D_h\|_{L^q(\mathbb{P})} \leq e^{2c_1^2}. \quad (47)$$

Recalling  $q = 2p$ , this is the first estimate in (42).

*Step 2: representation of  $D_h - 1$  as a stochastic integral.* The precise Itô calculation is as follows. Let

$$Y_t = M_t - \frac{1}{2}V_t, \quad D_t = e^{Y_t}.$$

The process  $V$  has finite variation, so  $[Y]_t = [M]_t = V_t$ . Itô's formula for the function  $y \mapsto e^y$  gives

$$\begin{aligned} dD_t &= D_t dY_t + \frac{1}{2}D_t d[Y]_t \\ &= D_t \left( dM_t - \frac{1}{2}dV_t \right) + \frac{1}{2}D_t dV_t \\ &= D_t dM_t = D_t \langle \theta_t, dB_t \rangle. \end{aligned}$$

Integrating this identity from zero to  $t$  gives the local-martingale representation

$$D_t - 1 = \int_0^t D_s \langle \theta_s, dB_s \rangle. \quad (48)$$

See [Karatzas and Shreve \[1991, Chapter 3\]](#) or [Revuz and Yor \[1999, Chapter IV\]](#) for Itô's formula and the quadratic variation rules used here.

*Step 3: application of BDG to an explicitly stopped martingale.* Although one can apply a local-martingale version of the Burkholder–Davis–Gundy inequality and then check integrability, it is clearer to localize first. For  $n \geq 2$ , define

$$\sigma_n = \tau_n \wedge \inf\{t \in [0, h] : D_t \geq n\}, \quad \inf \emptyset = \infty.$$

Both  $V$  and  $D$  are continuous and adapted, so  $\sigma_n$  is a stopping time. Stopping the process identity (48) and using (44) gives, for every deterministic  $t \in [0, h]$ ,

$$\begin{aligned} R_t^{(n)} &:= D_{t \wedge \sigma_n} - 1 = \int_0^{t \wedge \sigma_n} D_s \langle \theta_s, dB_s \rangle \\ &= \int_0^t \mathbf{1}_{\{s \leq \sigma_n\}} D_s \langle \theta_s, dB_s \rangle. \end{aligned}$$

Thus  $R^{(n)}$  is a square-integrable continuous martingale. Indeed, its quadratic variation is

$$[R^{(n)}]_h = \int_0^h \mathbf{1}_{\{s \leq \sigma_n\}} D_s^2 \|\theta_s\|^2 ds = \int_0^{h \wedge \sigma_n} D_s^2 \|\theta_s\|^2 ds \leq n^2 V_{h \wedge \tau_n} \leq n^3.$$

The possible distinction between  $s < \sigma_n$  and  $s \leq \sigma_n$  is immaterial for the Lebesgue and Itô integrals. For  $p \geq 2$ , apply the quantitative BDG inequality to the stopped continuous martingale  $R^{(n)}$  at the deterministic time  $h$ . In the explicit form of [Ren \[2008, Theorem 2\]](#),

$$\left\| R_h^{(n)} \right\|_{L^p(\mathbb{P})} \leq \left\| \sup_{0 \leq t \leq h} \left| R_t^{(n)} \right| \right\|_{L^p(\mathbb{P})} \leq 2\sqrt{2p} \left\| [R^{(n)}]_h^{1/2} \right\|_{L^p(\mathbb{P})}. \quad (49)$$

The resulting factor of order  $\sqrt{p}$  is the source of the final logarithmic loss. This is also the point at which the assumption  $p \geq 2$  is used. The choice  $q = 2p$  also ensures  $q > 1$ , as required for the  $L^q$  form of Doob's maximal inequality below.

Pathwise,

$$[R^{(n)}]_h^{1/2} \leq \left( \sup_{0 \leq s \leq h} D_s \right) V_h^{1/2}.$$

Hölder's inequality with exponents two and two therefore gives

$$\begin{aligned} \|[R^{(n)}]_h^{1/2}\|_{L^p(\mathbb{P})} &\leq \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h^{1/2}\|_{L^{2p}(\mathbb{P})} \\ &= \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h\|_{L^p(\mathbb{P})}^{1/2}. \end{aligned} \quad (50)$$

Because  $D$  is a nonnegative true martingale, Doob's maximal inequality [Revuz and Yor, 1999, Chapter II, Section 1] yields

$$\begin{aligned} \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} &\leq \frac{2p}{2p-1} \|D_h\|_{L^{2p}(\mathbb{P})} \\ &\leq C, \end{aligned}$$

where the last line uses (47) with  $q = 2p$ . Also, (38) gives

$$\|V_h\|_{L^p(\mathbb{P})}^{1/2} \leq CLh\sqrt{d+p}.$$

Substituting these two bounds into (49)–(50) gives, uniformly in  $n$ ,

$$\|D_{h \wedge \sigma_n} - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}. \quad (51)$$

To remove the stopping, note that the continuous functions  $t \mapsto V_t$  and  $t \mapsto D_t$  are bounded on the compact interval  $[0, h]$  for every sample path. Hence  $\sigma_n > h$  for all sufficiently large  $n$ , path by path, and  $D_{h \wedge \sigma_n} - 1 \rightarrow D_h - 1$  almost surely. Fatou's lemma applied to (51) proves

$$\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}.$$

This is the second estimate in (42) and completes the proof.  $\square$

### B.3 From path likelihood to MALA rejection

*Proof of Proposition B.1.* Choose the universal constant  $c$  in Proposition B.1 no larger than the constant in Lemma B.5 and sufficiently small that its step-size condition implies  $Lh < 1$ . Let  $S_h$  be the exact Langevin transition kernel. Define two endpoint probability measures on  $\mathbb{R}^d \times \mathbb{R}^d$ :

$$\nu_h(dx, dy) = \pi(dx)S_h(x, dy), \quad \mu_h(dx, dy) = \pi(dx)Q_h(x, dy).$$

The measure  $\nu_h$  is the law of  $(X_0, X_h)$  under the exact path law  $\mathbb{P}$ , whereas  $\mu_h$  is its law under the frozen-drift path law  $\mathbb{P}_h^{\text{fr}}$ . Because  $d\mathbb{P}_h^{\text{fr}}/d\mathbb{P} = D_h$  on  $\mathcal{F}_h$ , one has  $\mu_h \ll \nu_h$ . Let

$$F_h = \frac{d\mu_h}{d\nu_h}.$$

We next pass from the full-path likelihood  $D_h$  to the endpoint likelihood  $F_h$ . By the Doob–Dynkin lemma,  $\mathbb{E}[D_h \mid \sigma(X_0, X_h)]$  has a Borel-measurable representation as a function of the two endpoints; this function is a version of  $F_h$ . Indeed, for every bounded Borel function  $\varphi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ ,

$$\int \varphi(x, y) \mu_h(dx, dy) = \mathbb{E}_{\mathbb{P}_h^{\text{fr}}}[\varphi(X_0, X_h)]$$



$$\begin{aligned}
&= \mathbb{E}[D_h \varphi(X_0, X_h)] \\
&= \mathbb{E}\{\mathbb{E}[D_h \mid \sigma(X_0, X_h)] \varphi(X_0, X_h)\}.
\end{aligned}$$

This is exactly the defining Radon–Nikodym identity. Thus

$$F_h(X_0, X_h) = \mathbb{E}[D_h \mid \sigma(X_0, X_h)] \quad \mathbb{P}\text{-almost surely.} \quad (52)$$

This is the usual principle that the likelihood ratio of an observed statistic is the conditional expectation of the full-data likelihood ratio; see [Kallenberg \[2021\]](#) for conditional expectation, disintegration, and regular conditional distributions.

Subtracting one in (52) and using conditional Jensen’s inequality gives the  $L^p$ -contraction explicitly:

$$\begin{aligned}
\|F_h - 1\|_{L^p(\nu_h)}^p &= \mathbb{E}|\mathbb{E}[D_h - 1 \mid \sigma(X_0, X_h)]|^p \\
&\leq \mathbb{E}\{\mathbb{E}[|D_h - 1|^p \mid \sigma(X_0, X_h)]\} \\
&= \|D_h - 1\|_{L^p(\mathbb{P})}^p.
\end{aligned}$$

Hence

$$\|F_h - 1\|_{L^p(\nu_h)} \leq \|D_h - 1\|_{L^p(\mathbb{P})}. \quad (53)$$

We also need the conditional form of  $\mu_h = F_h \nu_h$ . Evaluating this measure identity on a rectangle  $A \times B$  gives

$$\int_A Q_h(x, B) \pi(\mathrm{d}x) = \int_A \int_B F_h(x, y) S_h(x, \mathrm{d}y) \pi(\mathrm{d}x). \quad (54)$$

Fix a countable  $\pi$ -system  $\mathcal{C}$  that generates the Borel  $\sigma$ -field on  $\mathbb{R}^d$ , for example, the finite intersections of open balls with rational centers and radii, together with  $\mathbb{R}^d$ . For each  $B \in \mathcal{C}$ , equality (54) for every Borel  $A$  implies equality of the two integrands for  $\pi$ -almost every  $x$ . Since  $\mathcal{C}$  is countable, there is one  $\pi$ -null set outside which this holds simultaneously for every  $B \in \mathcal{C}$ . For each fixed  $x$  outside that null set, both sides are measures in  $B$ , and the monotone-class theorem extends their equality from  $\mathcal{C}$  to all Borel sets. Thus, for  $\pi$ -almost every  $x$ ,

$$Q_h(x, \mathrm{d}y) = F_h(x, y) S_h(x, \mathrm{d}y). \quad (55)$$

Reversibility of exact Langevin makes  $\nu_h$  symmetric. Hence the transpose proposal flow  $\mu_h^\top(\mathrm{d}x, \mathrm{d}y) = \mu_h(\mathrm{d}y, \mathrm{d}x)$  has density  $F_h(y, x)$  relative to  $\nu_h$ . On the set where  $F_h(x, y) > 0$ , the Metropolis ratio is the Radon–Nikodym quotient  $F_h(y, x)/F_h(x, y)$ ; on the complementary set the usual zero-density convention gives the same formula below. Therefore the density of the accepted proposal flow relative to  $\nu_h$  is

$$F_h(x, y) \min\left\{1, \frac{F_h(y, x)}{F_h(x, y)}\right\} = \min\{F_h(x, y), F_h(y, x)\}.$$

Equivalently, the accepted flow is the meet of the proposal flow and its transpose [[Tierney, 1998](#), [Billera and Diaconis, 2001](#)]:

$$\pi(\mathrm{d}x) Q_h(x, \mathrm{d}y) \alpha_h(x, y) = \min\{F_h(x, y), F_h(y, x)\} \nu_h(\mathrm{d}x, \mathrm{d}y). \quad (56)$$

Disintegrate both sides of (56) with respect to the first coordinate, using the same countable-generator and monotone-class argument as in the proof of (55). Then (55) gives, for  $\pi$ -almost every  $x$ ,

$$r_h(x) = \int [F_h(x, y) - \min\{F_h(x, y), F_h(y, x)\}] S_h(x, \mathrm{d}y)$$

$$\leq \int (|F_h(x, y) - 1| + |F_h(y, x) - 1|) S_h(x, dy).$$

Define

$$A_h(x) = \int |F_h(x, y) - 1| S_h(x, dy), \quad B_h(x) = \int |F_h(y, x) - 1| S_h(x, dy).$$

Then  $r_h \leq A_h + B_h$ . Jensen's inequality for the probability measure  $S_h(x, dy)$  gives

$$\begin{aligned} \|A_h\|_{L^p(\pi)}^p &\leq \int |F_h(x, y) - 1|^p \nu_h(dx, dy), \\ \|B_h\|_{L^p(\pi)}^p &\leq \int |F_h(y, x) - 1|^p \nu_h(dx, dy). \end{aligned}$$

The first right-hand side is  $\|F_h - 1\|_{L^p(\nu_h)}^p$ ; the second is the same because  $\nu_h$  is symmetric. Minkowski's inequality therefore yields

$$\|r_h\|_{L^p(\pi)} \leq \|A_h\|_{L^p(\pi)} + \|B_h\|_{L^p(\pi)} \leq 2 \|F_h - 1\|_{L^p(\nu_h)}. \quad (57)$$

Combining (57), (53), and Lemma B.5 gives

$$\|r_h\|_{L^p(\pi)} \leq 2 \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh \sqrt{p(d+p)},$$

after absorbing the factor two into the universal constant. This is (29).  $\square$

## B.4 Proving Proposition 3.2

*Proof.* Decrease  $c_r$ , if necessary, so that  $c_r \leq 1$  and so that Proposition B.1 applies under (4). Fix  $p \geq 2$  and  $t$  satisfying that condition. Since  $d \geq 1$  and  $p \geq 2$ , it also implies  $t < 2/L$ . Define the rejection probability averaged over  $[t/2, t]$  by

$$\bar{r}_t(x) = \frac{2}{t} \int_{t/2}^t r_h(x) dh.$$

After decreasing the universal step constant, Proposition B.1 applies to every  $h \in [t/2, t]$ . Minkowski's integral inequality gives

$$\begin{aligned} \|\bar{r}_t\|_{L^p(\pi)} &\leq \frac{2}{t} \int_{t/2}^t \|r_h\|_{L^p(\pi)} dh \\ &\leq \frac{3C}{4} Lt \sqrt{p(d+p)}. \end{aligned}$$

Set

$$G_{t,p} = \{x : \bar{r}_t(x) \leq 1/3\}.$$

Markov's inequality yields

$$\pi(G_{t,p}^c) \leq \left( \frac{9C}{4} Lt \sqrt{p(d+p)} \right)^p,$$

which is (5) after renaming the universal constant.

We next compare nearby proposal laws. The Baillon–Haddad cocoercivity inequality states that [Nesterov, 2004, Theorem 2.1.5]

$$\langle \nabla U(x) - \nabla U(y), x - y \rangle \geq \frac{1}{L} \|\nabla U(x) - \nabla U(y)\|^2.$$

See also [Bauschke and Combettes \[2010\]](#). Consequently, for  $h \leq 2/L$ ,

$$\begin{aligned} & \|x - h\nabla U(x) - y + h\nabla U(y)\|^2 \\ &= \|x - y\|^2 - 2h \langle x - y, \nabla U(x) - \nabla U(y) \rangle + h^2 \|\nabla U(x) - \nabla U(y)\|^2 \\ &\leq \|x - y\|^2. \end{aligned}$$

Thus the proposal mean map is nonexpansive. Two Gaussian laws with covariance  $2hI_d$  and means  $\mu_x, \mu_y$  have Kullback–Leibler divergence  $\|\mu_x - \mu_y\|^2/(4h)$ . Pinsker’s inequality [[Tsybakov, 2009](#), Lemma 2.5] therefore gives

$$\text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \leq \frac{\|x - y\|}{\sqrt{8h}} \leq \frac{\|x - y\|}{2\sqrt{t}}, \quad h \in [t/2, t]. \quad (58)$$

For fixed  $x$ , write

$$R_{h,x}(\mathrm{d}y) = [1 - \alpha_h(x, y)] Q_h(x, \mathrm{d}y).$$

This measure has total mass  $r_h(x)$ , and, because  $Q_h(x, \cdot)$  has a Lebesgue density, it gives no mass to  $\{x\}$ . Hence

$$P_h(x, \cdot) - Q_h(x, \cdot) = r_h(x)\delta_x - R_{h,x}$$

is the Jordan decomposition into mutually singular positive and negative parts. With the convention  $\text{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$ , this proves

$$\text{TV}(P_h(x, \cdot), Q_h(x, \cdot)) = r_h(x). \quad (59)$$

Total variation is convex under mixing: for probability kernels  $M_h, N_h$  and a probability measure  $\rho$ ,

$$\text{TV}\left(\int M_h \mathrm{d}\rho, \int N_h \mathrm{d}\rho\right) \leq \int \text{TV}(M_h, N_h) \mathrm{d}\rho.$$

Applying this fact three times, together with the triangle inequality, (59), and (58), gives, for  $x, y \in G_{t,p}$  with  $\|x - y\| \leq \sqrt{t}/16$ ,

$$\begin{aligned} \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) &\leq \bar{r}_t(x) + \frac{2}{t} \int_{t/2}^t \text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \mathrm{d}h + \bar{r}_t(y) \\ &\leq \frac{1}{3} + \frac{1}{32} + \frac{1}{3} = \frac{67}{96} < \frac{3}{4}. \end{aligned}$$

This proves the first assertion of [Proposition 3.2](#).

It remains to prove the final assertion. Write  $g_x = \nabla U(x)$ , let  $Y = x - hg_x + \sqrt{2h}Z$ , and put  $g_Y = \nabla U(Y)$ . We first derive the log-ratio identity used below. Since

$$q_h(a, b) = (4\pi h)^{-d/2} \exp\left[-\frac{\|b - a + h\nabla U(a)\|^2}{4h}\right],$$

one has

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) \\ &\quad + \frac{1}{4h} \left( \|Y - x + hg_x\|^2 - \|x - Y + hg_Y\|^2 \right). \end{aligned}$$

Now  $Y - x + hg_x = \sqrt{2h}Z$  and  $x - Y = hg_x - \sqrt{2h}Z$ . Expanding the two squared norms and collecting the gradient and Gaussian terms gives

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) - \langle g_Y, x - Y \rangle - \frac{h}{4} \|g_Y - g_x\|^2 \\ &\quad - \sqrt{\frac{h}{2}} \langle Z, g_Y - g_x \rangle. \end{aligned} \tag{60}$$

The Baillon–Haddad inequality [Nesterov, 2004, Theorem 2.1.5] gives

$$U(x) - U(Y) - \langle g_Y, x - Y \rangle \geq \frac{1}{2L} \|g_Y - g_x\|^2.$$

If  $h \leq 1/L$ , the right-hand side of (60) is therefore at least

$$\frac{1}{4L} \|g_Y - g_x\|^2 - \sqrt{\frac{h}{2}} \|Z\| \|g_Y - g_x\| \geq -\frac{Lh}{2} \|Z\|^2.$$

It follows that

$$\alpha_h(x, Y) \geq e^{-Lh\|Z\|^2/2}, \quad \mathbb{E}\alpha_h(x, Y) \geq (1 + Lh)^{-d/2} \geq e^{-Lhd/2}.$$

Let  $t' \in (0, 1/(2Ld)]$ . Then every  $h \in [t'/2, t']$  has rejection probability at most  $1 - e^{-1/4}$ , uniformly in  $x$ . Combining two such rejection bounds with (58), we obtain, whenever  $\|x - y\| \leq \sqrt{t'}/16$ ,

$$\text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq 2(1 - e^{-1/4}) + \frac{1}{32} < \frac{3}{4}.$$

This proves the final assertion of Proposition 3.2 and completes the proof.  $\square$

## C Proof of Proposition 3.3

We repeatedly use the following notation for the standard Gaussian distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(v) dv, \quad \bar{\Phi}(x) = 1 - \Phi(x).$$

### C.1 Some basic properties of Gaussian distribution functions

Equation (6), due to Bakry and Ledoux [1996], states that

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{mr}).$$

We analyze the map  $(q, s) \mapsto \Phi(\Phi^{-1}(q) + s)$ .

We begin with a Mills-ratio bound.

**Lemma C.1.** *Let  $a \geq 0$ . Then*

$$\frac{\varphi(a)}{1+a} \leq \bar{\Phi}(a) \leq \frac{2\varphi(a)}{1+a}. \tag{61}$$

Moreover,

$$\log \frac{1}{\bar{\Phi}(a)} \leq (1+a)^2. \tag{62}$$

*Proof.* Let

$$g(a) = \bar{\Phi}(a) - \frac{\varphi(a)}{1+a}.$$

Then  $g(a) \rightarrow 0$  as  $a \rightarrow \infty$ , and direct differentiation gives

$$g'(a) = -\frac{a\varphi(a)}{(1+a)^2} \leq 0.$$

Thus  $g(a) \geq 0$ , proving the lower bound in (61). Similarly, for

$$f(a) = \frac{2\varphi(a)}{1+a} - \bar{\Phi}(a),$$

we have  $f(a) \rightarrow 0$  as  $a \rightarrow \infty$  and

$$f'(a) = -\frac{(1+a^2)\varphi(a)}{(1+a)^2} < 0.$$

Hence  $f(a) \geq 0$ , proving the upper bound. Classical sharper Mills bounds for  $a > 0$  go back at least to [Gordon \[1941\]](#).

The lower bound in (61) implies

$$\log \frac{1}{\bar{\Phi}(a)} \leq \frac{a^2}{2} + \log(1+a) + \frac{1}{2} \log(2\pi).$$

The difference between  $(1+a)^2$  and the right-hand side is positive at  $a = 0$ , and its derivative is

$$2+a - \frac{1}{1+a} > 0.$$

This proves (62). □

**Lemma C.2.** *Let  $q \in (0, 1/2]$  and  $s \geq 0$ . Then*

$$\Phi(\Phi^{-1}(q) + s) - q \geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}.$$

*Proof.* Write  $\Phi^{-1}(q) = -a$ , so  $a \geq 0$  and  $q = \bar{\Phi}(a)$ . Then

$$\Phi(-a+s) - \Phi(-a) = \int_0^s \varphi(a-v) dv.$$

For  $0 \leq v \leq \min\{s, 1\}$ ,

$$\frac{\varphi(a-v)}{\varphi(a)} = e^{av-v^2/2} \geq e^{-1/2}.$$

By the upper Mills bound in (61),  $\varphi(a) \geq q(1+a)/2$ . Hence

$$\begin{aligned} \Phi(-a+s) - \Phi(-a) &\geq \frac{e^{-1/2}}{2} q(1+a) \min\{s, 1\} \\ &\geq \frac{e^{-1/2}}{2} q \min\{1, s(1+a)\} \\ &\geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}. \end{aligned}$$

The second line uses  $(1+a) \min\{s, 1\} \geq \min\{1, s(1+a)\}$ ; the third uses (62) and  $e^{-1/2}/2 > 1/4$ . □

## C.2 Proving Proposition 3.3

*Proof.* If  $r = 0$ , the right-hand side of (7) is zero, so the conclusion is immediate. Assume  $r > 0$  and, without loss of generality, that  $\pi(A) = q$ . Since  $\text{dist}(A, B) \geq r$ , the open enlargement  $A^r$  is disjoint from  $B$ , and hence  $A^r \subseteq A \cup C$ . Combining (6) and Lemma C.2 gives

$$\begin{aligned}\pi(C) &\geq \pi(A^r) - \pi(A) \\ &\geq \Phi(\Phi^{-1}(q) + \sqrt{m}r) - q \\ &\geq \frac{q}{4} \min \left\{ 1, r\sqrt{m \log(1/q)} \right\}.\end{aligned}$$

This is exactly (7). □

## D Proof of Proposition 3.4

### D.1 A generic conductance bound

We begin with a generic one-step flow bound for kernels that satisfy a close-coupling condition under the Gaussian isoperimetric inequality in Proposition 3.3. Combining close coupling with an isoperimetric inequality to lower-bound one-step flows, and hence conductance, is a standard technique. Related defective- or truncated-conductance arguments appear in Dwivedi et al. [2019], Wu et al. [2022], Chen and Gatmiry [2023], Liu and Tong [2026].

**Lemma D.1.** *Let  $K$  be  $\pi$ -reversible, let  $t > 0$ , and let  $G \subseteq \mathbb{R}^d$  satisfy*

$$x, y \in G, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(K(x, \cdot), K(y, \cdot)) \leq \frac{3}{4}.$$

*Let  $S$  be a measurable set such that  $0 < \pi(S) \leq 1/2$ , and suppose that*

$$\pi(G^c) \leq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log(1/\pi(S))} \right\}. \quad (63)$$

*Then*

$$J_K(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log(1/\pi(S))} \right\}. \quad (64)$$

*Proof.* Fix a measurable  $S$  with  $0 < \pi(S) \leq 1/2$ . Let  $s = \pi(S)$  and

$$\beta = \min \left\{ 1, \sqrt{mt \log(1/s)} \right\}.$$

Suppose, for contradiction, that

$$J_K(S, S^c) < 2^{-13} s \beta.$$

Define

$$\begin{aligned}S_1 &= \{x \in S \cap G : K(x, S^c) < 1/16\}, \\ S_2 &= \{y \in S^c \cap G : K(y, S) < 1/16\}.\end{aligned}$$

For  $x \in S_1$  and  $y \in S_2$ ,

$$\text{TV}(K(x, \cdot), K(y, \cdot)) \geq K(x, S) - K(y, S) > \frac{7}{8}.$$

The close-coupling condition therefore implies

$$\text{dist}(S_1, S_2) \geq \frac{\sqrt{t}}{16}. \quad (65)$$

Markov's inequality and reversibility give

$$\begin{aligned} \pi(S_1) &\geq s - \pi(S \cap G^c) - 16J_K(S, S^c), \\ \pi(S_2) &\geq 1 - s - \pi(S^c \cap G^c) - 16J_K(S, S^c). \end{aligned}$$

By (63) and the assumed strict upper bound on the flow,

$$\pi(S_1) > s - 17 \cdot 2^{-13} s \beta, \quad \pi(S_2) > 1 - s - 17 \cdot 2^{-13} s \beta.$$

Since  $0 < \beta \leq 1$ ,  $17 \cdot 2^{-13} < 1/2$ , and  $1 - s \geq s$ , it follows that

$$\pi(S_1) \geq \frac{s}{2}, \quad \pi(S_2) \geq \frac{s}{2}. \quad (66)$$

Set  $q' = \min\{\pi(S_1), \pi(S_2)\}$ . Because  $S_1 \subseteq S$ ,

$$\frac{s}{2} \leq q' \leq s \leq \frac{1}{2}.$$

Apply [Proposition 3.3](#) to  $S_1, S_2$ , and their complement. Since  $q' \leq s$ , one has  $\log(1/q') \geq \log(1/s)$ . Using (65), (66), and  $\min\{1, x/16\} \geq \min\{1, x\}/16$ , we obtain

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &\geq \frac{q'}{4} \min \left\{ 1, \frac{1}{16} \sqrt{mt \log(1/q')} \right\} \\ &\geq \frac{s}{8} \cdot \frac{1}{16} \min \left\{ 1, \sqrt{mt \log(1/s)} \right\} = \frac{s}{128} \beta. \end{aligned} \quad (67)$$

On the other hand, the parts removed from  $S$  and  $S^c$  are disjoint, so we have the sharper upper bound

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &\leq \pi(G^c) + 32J_K(S, S^c) \\ &< 33 \cdot 2^{-13} s \beta < \frac{s}{128} \beta, \end{aligned}$$

contradicting (67). Therefore the assumed strict upper bound on the flow is impossible, and (64) holds.  $\square$

## D.2 A technical lemma

The proof of [Proposition 3.4](#) via [Lemma D.1](#) requires one additional technical lemma.

**Lemma D.2.** *Let  $c_r, C_r$  be the constants in [Proposition 3.2](#). Fix a number  $b_0 > 0$  satisfying*

$$b_0 \leq \min \left\{ \frac{1}{2}, c_r, \frac{1}{16C_r} \right\}. \quad (68)$$

*Let  $p \geq p_\star := A_0\{1 + \log(d+1) + \log \kappa\}$ ,  $\theta \in (0, 1]$ , and*

$$t = \frac{\theta b_0}{L \sqrt{p(d+p)}},$$

where  $A_0$  satisfies

$$A_0 \geq \max \left\{ 2, \frac{13 \log 2 + \frac{1}{2} \log(2/b_0) + 1}{\log 16 - \frac{1}{2}} \right\}. \quad (69)$$

Then, for every  $u \in [\log 2, p/2]$ ,

$$\left( \frac{\theta}{16} \right)^p \leq 2^{-13} e^{-u} \sqrt{mtu}. \quad (70)$$

Moreover,  $mtu \leq b_0/2 < 1$  throughout this range.

*Proof.* The last assertion follows directly from  $m/L = 1/\kappa$ :

$$mtu = \frac{\theta b_0 u}{\kappa \sqrt{p(d+p)}} \leq \frac{\theta b_0}{2\kappa} \sqrt{\frac{p}{d+p}} \leq \frac{b_0}{2}.$$

We focus on proving (70). Since  $u \mapsto e^{-u} \sqrt{u}$  decreases on  $[1/2, \infty)$ , the right-hand side of (70) is smallest at  $u = p/2$ . At that endpoint it equals

$$2^{-13} e^{-p/2} \left( \frac{\theta b_0}{2\kappa} \right)^{1/2} \left( \frac{p}{d+p} \right)^{1/4}.$$

Dividing the desired inequality at  $u = p/2$  by  $\theta^{1/2}$  shows that its left-hand side contains the factor  $\theta^{p-1/2}$ . Since  $p \geq 2$  and  $0 < \theta \leq 1$ , this factor is at most one. It is therefore sufficient that

$$p \left( \log 16 - \frac{1}{2} \right) \geq 13 \log 2 + \frac{1}{2} \log \frac{2\kappa}{b_0} + \frac{1}{4} \log \frac{d+p}{p}. \quad (71)$$

Since  $p \geq 1$ ,

$$\log \frac{d+p}{p} \leq \log(d+1).$$

The right-hand side of (71) is therefore at most

$$13 \log 2 + \frac{1}{2} \log \frac{2}{b_0} + \frac{1}{2} \log \kappa + \frac{1}{4} \log(d+1).$$

Thus any  $A_0$  satisfying (69) is sufficient. Indeed, this choice gives

$$A_0 \left( \log 16 - \frac{1}{2} \right) \geq 13 \log 2 + \frac{1}{2} \log(2/b_0) \quad \text{and} \quad A_0 \left( \log 16 - \frac{1}{2} \right) \geq 1 > \frac{1}{2}.$$

Since  $p \geq p_\star = A_0 \{1 + \log(d+1) + \log \kappa\}$ , multiplying these coefficient-wise bounds by 1,  $\log(d+1)$ , and  $\log \kappa$  proves (71), uniformly in all parameters.  $\square$

### D.3 Proving Proposition 3.4

*Proof.* We begin with the first assertion. Let  $A_0$  and  $b_0$  be universal constants satisfying (68) and (69). Let  $p \geq p_\star := A_0 \{1 + \log(d+1) + \log \kappa\}$ ,  $\theta \in (0, 1]$ , and

$$t = \frac{\theta b_0}{L \sqrt{p(d+p)}}.$$

Let  $S \subseteq \mathbb{R}^d$  be measurable with  $e^{-p/2} \leq \pi(S) \leq 1/2$ . By Lemma D.2, since  $\log(1/\pi(S)) \in [\log 2, p/2]$ ,

$$\left( \frac{\theta}{16} \right)^p \leq 2^{-13} \pi(S) \sqrt{mt \log(1/\pi(S))} = 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log(1/\pi(S))} \right\}.$$



Because  $b_0 \leq c_r$  and  $b_0 \leq 1/(16C_r)$ , the first assertion of [Proposition 3.2](#) yields a measurable set  $G_{t,p} \subseteq \mathbb{R}^d$  such that

$$\pi(G_{t,p}^c) \leq \left[ C_r L \frac{\theta b_0}{L \sqrt{p(d+p)}} \sqrt{p(d+p)} \right]^p \leq \left( \frac{\theta}{16} \right)^p,$$

and

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

Then, by [Lemma D.1](#),

$$J_{\mathcal{K}_t}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log(1/\pi(S))} \right\} = 2^{-13} \pi(S) \sqrt{mt \log(1/\pi(S))}.$$

The equality follows from the final assertion of [Lemma D.2](#), namely,  $mt \log(1/\pi(S)) \leq 1$ . This proves the first assertion.

For the second assertion, let  $t' \in (0, 1/(2Ld)]$  and let  $S \subseteq \mathbb{R}^d$  be measurable with  $0 < \pi(S) \leq 1/2$ . By the second assertion of [Proposition 3.2](#),

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

Applying [Lemma D.1](#) with  $G = \mathbb{R}^d$  yields

$$J_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt' \log(1/\pi(S))} \right\}.$$

This proves the second assertion. □

## E Proof of Lemma 3.5

The result can also be derived from the general Cheeger inequality of [Chen and Wang \[2000\]](#). The proof below is direct and self-contained; we include the details because the precise dependence on  $\beta_j$  and  $\gamma_j$  is needed later.

*Proof.* For  $j = 1, \dots, N$ , define the symmetric measure

$$J_j(dx, dy) = \varpi(dx) K_j(x, dy).$$

The measure  $J_j$  is symmetric because  $K_j$  is  $\varpi$ -reversible. Both of its marginals equal  $\varpi$ .

We first prove the result for a nonnegative function  $g \in L^2(\varpi)$  such that  $\varpi(g > 0) \leq 1/2$ . If  $\sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) = \infty$ , the desired inequality is immediate, so assume this sum is finite. For  $t \geq 0$ , let

$$A_t = \{x : g(x)^2 > t\}.$$

Then  $\varpi(A_t) \leq 1/2$ , so the assumed fractional boundary inequality gives

$$\varpi(A_t) \leq \sum_{j=1}^N \beta_j J_{K_j}(A_t, A_t^c).$$

Integrating this inequality over  $t \geq 0$  and using the layer-cake identity yields

$$\begin{aligned}\|g\|_{L^2(\varpi)}^2 &= \int_0^\infty \varpi(A_t) dt \\ &\leq \sum_{j=1}^N \beta_j \int_0^\infty J_{K_j}(A_t, A_t^c) dt.\end{aligned}$$

For each  $j$ , symmetry of  $J_j$  gives the coarea identity

$$\int_0^\infty J_{K_j}(A_t, A_t^c) dt = \frac{1}{2} \iint |g(x)^2 - g(y)^2| J_j(dx, dy).$$

Indeed, for fixed  $x, y$ ,

$$\int_0^\infty \mathbf{1}_{\{g(x)^2 > t \geq g(y)^2\}} dt = [g(x)^2 - g(y)^2]_+,$$

and symmetry changes the integral of the positive part into one half of the integral of the absolute difference. Consequently,

$$\|g\|_{L^2(\varpi)}^2 \leq \frac{1}{2} \sum_{j=1}^N \beta_j \iint |g(x) - g(y)| |g(x) + g(y)| J_j(dx, dy).$$

Apply Cauchy–Schwarz simultaneously over the component index  $j$  and the pair  $(x, y)$ :

$$\begin{aligned}\|g\|_{L^2(\varpi)}^2 &\leq \frac{1}{2} \left\{ \sum_{j=1}^N \gamma_j \iint [g(x) - g(y)]^2 J_j(dx, dy) \right\}^{1/2} \\ &\quad \times \left\{ \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \iint [g(x) + g(y)]^2 J_j(dx, dy) \right\}^{1/2}.\end{aligned}$$

The first square root equals

$$\left[ 2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2}.$$

For the second square root, the inequality

$$[g(x) + g(y)]^2 \leq 2g(x)^2 + 2g(y)^2$$

and the fact that both marginals of  $J_j$  equal  $\varpi$  give

$$\iint [g(x) + g(y)]^2 J_j(dx, dy) \leq 4 \|g\|_{L^2(\varpi)}^2.$$

It follows that

$$\|g\|_{L^2(\varpi)}^2 \leq \left[ 2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2} \|g\|_{L^2(\varpi)} \left[ \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right]^{1/2}.$$

If  $g = 0$  in  $L^2(\varpi)$ , there is nothing to prove. Otherwise, cancelling one factor of  $\|g\|_{L^2(\varpi)}$  and squaring gives

$$\sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \geq \frac{\|g\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}. \quad (72)$$

Now let  $f \in L^2(\varpi)$ , and let  $c$  be a median of  $f$ ; thus

$$\varpi(f \geq c) \geq \frac{1}{2}, \quad \varpi(f \leq c) \geq \frac{1}{2}.$$

Define the positive and negative parts of  $f - c$ :

$$g_+ = (f - c)_+, \quad g_- = (f - c)_-.$$

Then

$$\varpi(g_+ > 0) \leq \frac{1}{2}, \quad \varpi(g_- > 0) \leq \frac{1}{2}.$$

Moreover, for every  $x, y$ ,

$$\begin{aligned} [f(x) - f(y)]^2 &\geq [g_+(x) - g_+(y)]^2 \\ &\quad + [g_-(x) - g_-(y)]^2. \end{aligned}$$

To see this, write

$$f - c = g_+ - g_-.$$

The two differences

$$g_+(x) - g_+(y) \quad \text{and} \quad g_-(x) - g_-(y)$$

have opposite signs, so the cross term in the square of their difference is nonnegative.

Therefore, by (72),

$$\begin{aligned} \mathcal{E}_P(f, f) &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \\ &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_+, g_+) + \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_-, g_-) \\ &\geq \frac{\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}. \end{aligned}$$

Since  $g_+$  and  $g_-$  have disjoint supports and  $f - c = g_+ - g_-$ ,

$$\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2 = \|f - c\|_{L^2(\varpi)}^2.$$

The mean minimizes  $a \mapsto \|f - a\|_{L^2(\varpi)}^2$ , so

$$\|f - c\|_{L^2(\varpi)}^2 \geq \text{Var}_\varpi(f).$$

We conclude that

$$\mathcal{E}_P(f, f) \geq \frac{\text{Var}_\varpi(f)}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}.$$

Taking the infimum over all nonconstant  $f \in L^2(\varpi)$  proves the claim.  $\square$

## F Multiscale arithmetic

This appendix verifies the summation estimate used in [Section 4](#). The constants  $b_0$  and  $A_0$  are the universal choices made at the beginning of the proof of [Theorem 2.1](#).

**Lemma F.1.** *Assume  $p_\star < d$ , and define  $p_j, J, t_j, \theta$  by (8)–(10). With*

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-26} m t_0 \log 2, \quad \phi_j^2 = 2^{-29} m t_j p_j \quad (j \geq 1),$$

*there is a universal  $C > 0$  such that*

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{H L^2 p_\star (d + p_\star)}{m \theta^2 b_0^2}.$$

*Proof.* For the first component,

$$\frac{1}{\gamma_0 \phi_0^2} = \frac{2^{27} H}{m t_0^2 \log 2} = \frac{2^{27} H L^2 p_\star (d + p_\star)}{m \theta^2 b_0^2 \log 2}.$$

For  $j \geq 1$ ,

$$\frac{1}{\gamma_j \phi_j^2} = \frac{2^{30} H}{m t_j^2 p_j} = \frac{2^{30} H L^2 (d + p_j)}{m \theta^2 b_0^2}.$$

It remains to bound the sum of  $d + p_j$ . Because  $p_\star < d$ , one has  $J \geq 1$ . Minimality of  $J$  gives  $4^{J-1} p_\star < d$ , and hence

$$J < 1 + \log_4 \frac{d}{p_\star} \leq 1 + \log_4 d.$$

The geometric-series formula and (9) give

$$\sum_{j=1}^J p_j = p_\star \sum_{j=1}^J 4^j = \frac{4p_J - 4p_\star}{3} < \frac{16d}{3}.$$

Moreover, because  $p_\star \geq A_0[1 + \log(d + 1)]$ , it holds that

$$1 + \log_4 d \leq 1 + \frac{p_\star}{A_0 \log 4} \leq C p_\star,$$

where  $C$  is universal because  $A_0 \geq 2$  and  $p_\star \geq 2$ . Therefore

$$\sum_{j=1}^J (d + p_j) = Jd + \sum_{j=1}^J p_j \leq C p_\star d.$$

Adding the first-component term and using  $p_\star \geq 1$  yields

$$p_\star (d + p_\star) + \sum_{j=1}^J (d + p_j) \leq C p_\star (d + p_\star),$$

which proves the claimed estimate. □

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